NON-INVASIVE MONITORING OF VITAL SIGNS FOR BEDRIDDEN PATIENT

NOR MAISARAH BINTI ISMAIL 2213080

DEPARTMENT OF MECHATRONICS ENGINEERING

KULLIYAH OF ENGINEEERING

INTERNATIONAL ISLAMIC UNIVERSITY MALAYSIA

JULY 2026

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NOR MAISARAH BINTI ISMAIL 2213080

PROJECT SUPERVISOR: DR. NUR LIYANA BINTI AZMI

A REPORT SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENT FOR A DEGREE OF BACHELOR OF MECHATRONICS ENGINEERING WITH HONOURS

ABSTRACT

Continuous heart-rate monitoring is important for assessing the health of bedridden patients. However, conventional electrocardiogram (ECG) devices require electrodes to be attached directly to the skin, which may cause discomfort during prolonged use. This project aimed to develop a non-invasive sensor pad for detecting cardiac-related mechanical activity, extract heart-rate information from the captured signals and evaluate its accuracy against an ECG reference device. The sensor pad consisted of three FSR 408 strip sensors placed on top of the mattress beneath the subject's thoracic region. Four subjects were tested in the supine position. The FSR signals were acquired using an Arduino Uno, while reference ECG signals were recorded simultaneously using a BITalino device at 100 Hz. The recordings were synchronised, trimmed, filtered and normalised in MATLAB before peak detection and heart-rate estimation were performed. The best-performing sensors for Subjects 1, 2, 3 and 4 produced absolute heart-rate errors of 4.74, 0.87, 1.68 and 0.73 bpm, respectively. Sensor 1 showed the strongest overall performance, with a mean absolute error of 4.87 bpm and a mean matched-beat percentage of 50.27%. However, the best-performing sensor differed among the subjects, supporting the use of a multi-sensor array. Overall, the results demonstrate the feasibility of using an on mattress FSR sensor pad for non-invasive heart-rate monitoring. Future improvements should include adaptive peak detection, motion-artifact reduction, enhanced signal-conditioning circuitry and evaluation involving more subjects, different mattresses and additional body positions.

ACKNOWLEDGEMENTS

In the name of Allah, the Most Gracious and the Most Merciful.

First and foremost, I would like to express my deepest gratitude to Allah SWT for His endless blessings, guidance, and strength throughout the completion of this Final Year Project. Without His mercy and wisdom, this work would not have been possible.

I would like to express my sincere appreciation to my project supervisor, **Dr. Nur Liyana Binti Azmi**, for her invaluable guidance, continuous encouragement, constructive comment and unwavering support throughout the development of this project. Her expertise and patience have greatly contributed to the successful completion of this study.

My heartfelt thanks also go to the lecturers and staff of the Department of Mechatronics Engineering, International Islamic University Malaysia (IIUM) for providing the knowledge, facilities and assistance needed during the course of this research.

I would also like to extend my gratitude to my family for their endless love, prayers, understanding and moral support. Their encouragement has been my greatest motivation in overcoming every challenge encountered during this project.

Finally, I would like to thank my friends and classmates who have volunteered to be part of my project and provided valuable discussions, suggestions and encouragement throughout this journey. Their support and companionship have made the completion of this project a meaningful and memorable experience.

May Allah SWT bless everyone who has contributed directly or indirectly to the success of this project.

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LIST OF ABBREVIATIONS

ADC	Analog-to-Digital Converter
BCG	Ballistocardiogram
BPM	Beats per Minute
ECG	Electrocardiogram
FBG	Fibre Bragg Grating
FSR	Force-Sensitive Resistor
HR	Heart Rate
PSG	Polysomnography
PTT	Pulse Transit Time
PVDF	Polyvinylidene Fluoride
RR	Respiratory Rate
SD	Standard Deviation

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF STUDY

Vital signs are important physiological parameters used to assess a person's health condition. Among these parameters, heart rate (HR) provides valuable information about cardiovascular function and a patient's general condition. Continuous heart-rate monitoring supports the early detection of cardiovascular abnormalities, the assessment of patient recovery, and long-term healthcare management. Such monitoring is particularly important for bedridden patients who require continuous observation in hospitals, nursing homes or home-care environments.

Conventional heart-rate monitoring systems, such as electrocardiograms (ECGs) and pulse oximeters, generally require electrodes or sensors to be attached directly to the patient's body. Although these devices provide accurate measurements, prolonged use may cause discomfort, skin irritation, and restrictions on natural movement during rest or sleep. Furthermore, the repeated attachment and removal of sensors may increase the workload of healthcare personnel and reduce the practicality of continuous long-term monitoring, particularly for bedridden patients.

Recent advances in biomedical sensing technology have led to the development of less obtrusive monitoring methods based on ballistocardiography (BCG). BCG measures the small mechanical vibrations generated by cardiac contractions and the movement of blood through the body. These vibrations can be detected using sensors integrated into a bed or sensor pad, allowing cardiac activity to be monitored without attaching sensors directly to the patient's

skin. This approach may improve patient comfort while allowing the patient to remain in a natural lying position.

Different sensing technologies, including force-sensitive resistors (FSRs), load cells, and piezoelectric sensors have been investigated for bed-based cardiac monitoring because of their ability to detect small mechanical pressure variations. In this study, FSR 408 strip sensors were selected because they are thin, flexible, relatively inexpensive and provide a large sensing area. When combined with appropriate signal conditioning and digital signal-processing techniques, the pressure variations detected by the FSR sensors can be processed to estimate heart rate. Compared with more complex technologies, such as Fibre Bragg grating (FBG) sensors, an FSR-based system may provide a simpler and more cost-effective solution for bed-based monitoring applications.

Therefore, this study proposes the development of a non-invasive bed-based monitoring system for measuring the heart rate of bedridden patients using FSR sensor technology. The proposed system is intended to provide a comfortable and cost-effective method for monitoring heart rate without attaching sensors directly to the patient's skin. Its successful implementation could support continuous patient monitoring in hospitals, nursing homes and home-care environments while improving patient comfort and reducing dependence on conventional body-attached monitoring devices.

1.2 PROBLEM STATEMENT

Bedridden patients may require continuous heart-rate monitoring because changes in heart rate can provide important information about their cardiovascular condition and overall health. Conventional monitoring devices, such as electrocardiogram (ECG) machines and pulse

oximeters, require electrodes or sensors to be attached directly to the patient's body. Although these devices provide accurate measurements, prolonged use may cause discomfort, skin irritation, and inconvenience, particularly during rest or sleep. Bed-based monitoring systems provide a possible alternative by detecting the small mechanical signals generated by cardiac activity without attaching sensors directly to the patient's skin. However, signal quality may be affected by sensor placement, body movement, body characteristics, clothing material, mattress properties and external noise making consistent cardiac-peak detection and reliable heart-rate estimation challenging. Therefore, this study aims to develop and evaluate a low-cost, non-invasive bed-based monitoring system using an FSR-based sensor pad, appropriate signal-processing techniques, and an ECG reference device to measure the heart rate of bedridden patients.

1.3 RESEARCH OBJECTIVES

The aims of this research to develop a reliable, unobtrusive, and cost-effective monitoring system capable of tracking vital signs without direct contact. The specific objectives of the study are:

1. To design and develop a non-invasive sensor pad capable of detecting for bedridden patients.
2. To analyze signal conditioning and processing to extract heart rate data from signals captured by the sensor.
3. To evaluate the accuracy of the system by comparing measured vital sign with standard medical devices.

1.4 SUMMARY

This chapter introduced the concept of non-contact bed-based monitoring of vital signs and highlighted its significance in modern healthcare applications. The background of the study discussed the increasing demand for continuous and comfortable monitoring of heart rate and respiratory rate, particularly for elderly patients and individuals with chronic cardiovascular or respiratory conditions. It also emphasized the limitations of conventional contact-based monitoring devices which often cause discomfort and restrict patient movement during long-term use.

The problem statement identified the need for an unobtrusive and cost-effective monitoring system that can continuously measure vital signs without requiring direct skin contact. In response to this issue, the research aims to develop a bed-based monitoring system capable of detecting heart rate and respiratory rate using non-contact sensing technology and appropriate signal processing techniques.

Finally, the research objectives were presented to guide the study, including the design of a bed-based sensor system and the implementation of signal processing methods for vital sign extraction and the evaluation of the system's performance against standard medical devices. The following chapter reviews existing literature and related technologies that provide the theoretical foundation for the proposed monitoring system.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

The purpose of this study is to discover a new way to get the monitoring of the vital signs such as heart rate (HR). In this chapter, the review of past research paper and literature related to this study will be discussed. The review has been organized to see the difference between the conventional method and the improvised method to see the upgrading version which make the monitoring easier. This review can help to provides theoretical foundation and understanding to create this non-contact bed-based monitoring of vital signs.

2.2 CONVENTIONAL METHOD

Electrocardiography (ECG) is a conventional method used to measure the electrical activity of the heart. The ECG waveform consists of several components, including the P wave, QRS complex and T wave. The R-peak is the most prominent feature of the QRS complex and represents ventricular depolarisation. The time interval between consecutive R-peaks can be used to calculate heart rate.

Conventional ECG monitoring requires electrodes to be attached directly to the patient's skin. According to Ramasamy et al. (2018), these electrodes detect small electrical potential differences generated by cardiac activity and transmit the signals to a monitoring device. This method provides reliable cardiac measurements and is widely used in clinical environments. In the present study, ECG was therefore selected as the reference measurement for evaluating the heart rate estimated using the FSR sensor pad.

Despite its reliability, conventional ECG monitoring may be unsuitable for prolonged and unobtrusive monitoring. The electrodes, adhesive materials and connecting wires may cause discomfort, restrict natural movement and produce skin irritation during extended use. Signal quality may also be affected by poor electrode contact and body movement. These limitations support the development of a non-invasive bed-based monitoring system capable of detecting cardiac activity without attaching the primary sensing device directly to the patient's body.

2.3 BALLISTOCARDIOGRAM (BCG)

Ballistocardiography (BCG) is a non-invasive method used to measure small mechanical movements of the body caused by cardiac activity. These movements are generated when blood is ejected from the heart and circulated through the major blood vessels. Unlike electrocardiography (ECG), which records the electrical activity of the heart using electrodes attached to the skin, BCG detects the mechanical response produced by each heartbeat. According to Haghi et al. (2023) and Xu et al. (2020), these mechanical movements can be transmitted through the body and detected using sensors placed on or beneath a mattress.

A typical BCG waveform contains several components associated with different stages of the cardiac cycle. However, its waveform shape and amplitude may vary depending on the sensing technology, sensor placement, body position, mattress characteristics and pressure distribution. Cardiac-related peaks detected from the BCG signal can be used to identify individual heartbeats, and the intervals between consecutive peaks can be used to estimate heart rate (Pino et al., 2017).

Figure 2.1, obtained from Pino et al. (2017), illustrates the relationship between the ECG and BCG waveforms. The ECG R-peak represents the electrical activation associated with

ventricular contraction, while the corresponding BCG peak occurs later because it represents the resulting mechanical movement. This time delay allows the ECG signal to serve as a reference for identifying and validating cardiac-related peaks detected by bed-based mechanical sensors.

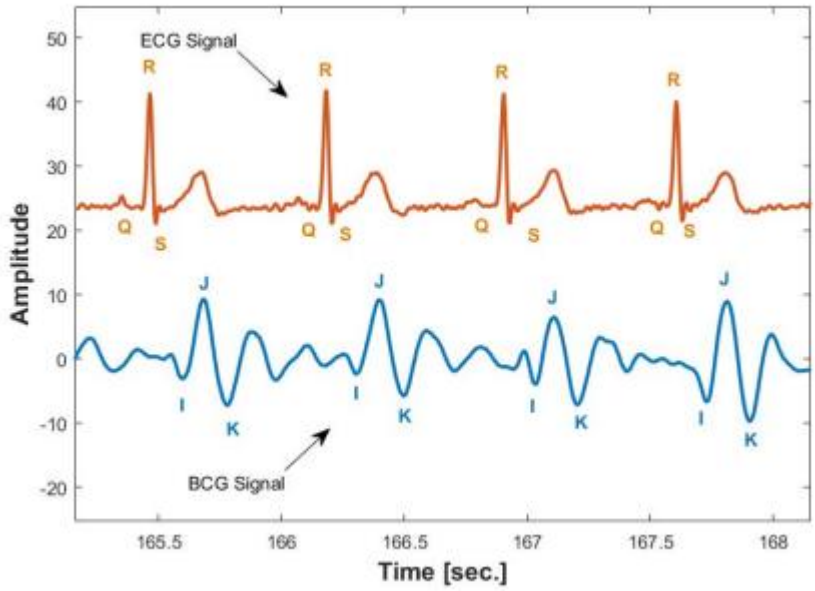


Figure 2.1 Comparison between ECG and BCG waveforms (Pino et al., 2017)

The ability to detect BCG signals without attaching sensors directly to the patient makes this approach suitable for unobtrusive bed-based monitoring. Various sensing technologies, including piezoelectric sensors, load sensors and force-sensitive resistors, have therefore been investigated for detecting cardiac-induced mechanical activity through a mattress. These technologies are discussed in the following section.

2.4 BED-BASED SENSING TECHNOLOGIES

Several sensing technologies have been developed to measure cardiac-related mechanical movements from a person lying on a bed. These technologies include fibre-optic sensors, piezoelectric sensors, load cells and force-sensitive resistors. Although they operate according to different sensing principles, they share the aim of obtaining physiological information

without attaching sensors directly to the patient's skin. Their suitability for continuous monitoring depends on factors such as sensitivity, installation method, equipment complexity, sensor placement and resistance to motion artefacts.

2.4.1 Fiber Optic Sensors

Fibre-optic sensing is one of the methods used for unobtrusive physiological monitoring. A commonly used fibre-optic sensor is the Fibre Bragg Grating (FBG), which contains a periodic structure within an optical fibre. When force, strain or deformation is applied to the fibre, the wavelength of the reflected light changes. These wavelength changes can be measured using an optical interrogator and analysed to detect small mechanical movements produced by cardiac and respiratory activity. Fibre-optic sensors offer high sensitivity and immunity to electromagnetic interference because their operation is based on light rather than electrical signals.

Zhu et al. (2015) developed an FBG sensor mat consisting of three sensor arrays, with six sensors in each array, resulting in a total of 18 sensing points. The arrays were positioned beneath the pillow, upper chest and lower chest areas to capture physiological movements from different locations. The arrangement of the FBG sensor arrays and optical interrogation system developed by Zhu et al. (2015) is illustrated in Figure 2.

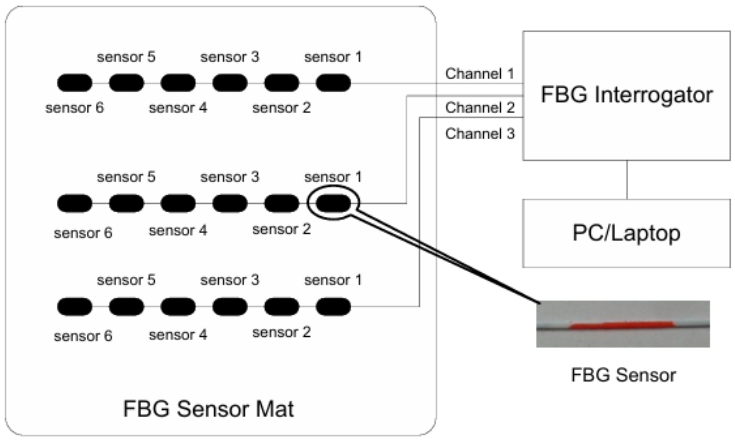


Figure 2.2 FBG-based bed monitoring system (Zhu et al., 2015)

The use of several sensing points improved signal availability because the system could select or combine signals obtained from different areas. The researchers applied spectrum and cepstrum analyses with multiple time-window lengths to assess signal quality and estimate respiratory rate. The system was evaluated using a polysomnography (PSG) respiratory belt as the reference measurement.

Although Zhu et al. (2015) primarily focused on respiratory-rate estimation, the study demonstrated that mechanical physiological signals could be acquired using a distributed sensor array integrated into a bed. However, an FBG-based system requires an optical interrogator to transmit and analyse wavelength information. The use of multiple optical sensors and specialised equipment may increase the system's cost and complexity. Therefore, although fibre-optic sensors are sensitive and resistant to electromagnetic interference, they may be less suitable for developing a simple and economical bed-based heart-rate monitoring system.

2.4.2 Piezoelectric Sensor

Piezoelectric sensors generate an electrical response when mechanical force or deformation is applied. Polyvinylidene fluoride (PVDF) film is commonly used for bed-based monitoring because it is thin, flexible and sensitive to small mechanical vibrations. A PVDF sensor placed beneath a mattress can therefore detect BCG activity without direct contact with the subject.

Xu et al. (2020) developed an unconstrained vital-sign monitoring system using a comb-shaped PVDF thin-film sensor measuring, as shown in Figure 2.3. The large sensing area allowed mechanical activity to be captured from different body locations. The system architecture, including its sensing, signal-conditioning and data-processing components, is presented in Figure 2.4.

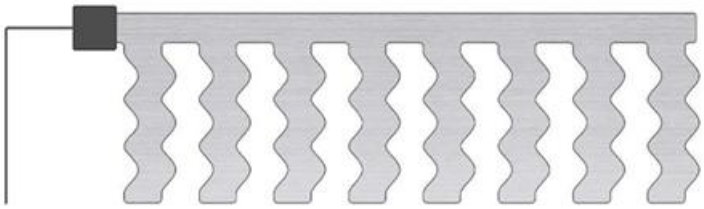


Figure 2.3 Comb-shaped PVDF thin-film mattress (Xu et al., 2020)

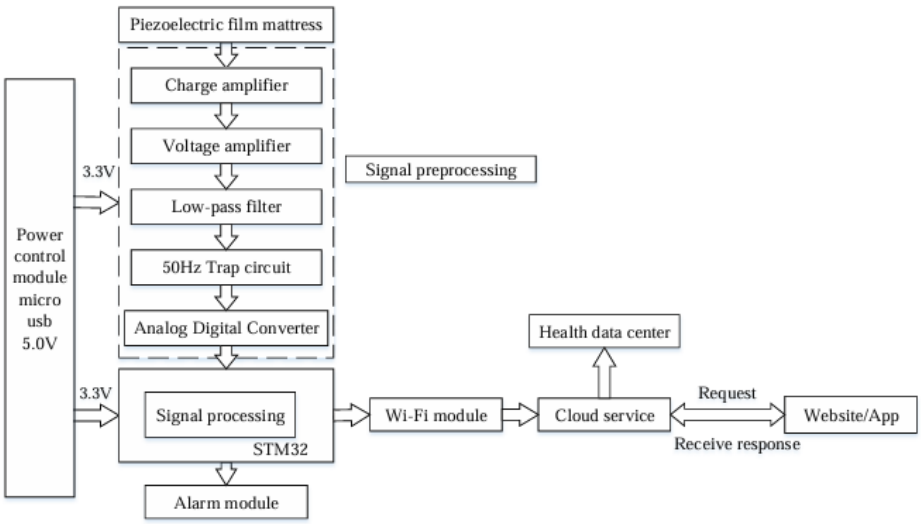


Figure 2.4 Architecture of the PVDF-based vital-sign monitoring mattress (Xu et al., 2020)

The study demonstrated that PVDF sensors could detect cardiac-related mechanical activity at a sampling frequency of 100 Hz. However, these sensors require amplification and signal-conditioning circuits, and their output may be influenced by body movement, sensor placement and mechanical coupling with the mattress.

2.4.3 LOAD-CELL SENSORS

Load sensors can be installed beneath the bed supports to detect changes in force caused by respiration, cardiac activity and body movement. This placement allows physiological activity to be monitored without modifying the mattress or attaching sensors to the patient.

Nishimura et al. (2025) developed a bed-based vital-sign monitoring system using four load cells installed beneath the four legs of a bed. The placement of the load sensors beneath the bed supports is shown in Figure 2.5. The load signals from the four corners were used to calculate changes in the displacement of the body's centre of mass. The recorded signals were processed using a Chebyshev low-pass filter and principal component analysis to isolate fluctuations associated with respiration. The system was evaluated in different sleeping postures to investigate the effect of changes in body position on the measurements.

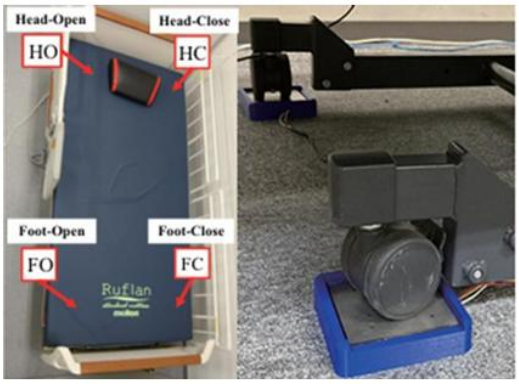


Figure 2.5 Placement of load sensors beneath the bed supports (Nishimura et al., 2025)

The study by Nishimura et al. (2025) focused mainly on respiratory-rate estimation rather than heart-rate measurement. However, it demonstrates an alternative method of integrating force sensors into an existing bed without attaching sensors to the patient. A limitation of this approach is that the measurements represent changes in the overall load distribution of the bed. Consequently, changes in posture and body movement may produce substantial variations in the recorded signals. The installation of a load cell beneath every bed leg may also be less convenient for a portable sensor-pad design.

2.4.4 Force-Sensitive Resistors (FSR)

Force-sensitive resistors (FSRs) are piezoresistive sensors whose electrical resistance decreases as the applied force increases. When used in a voltage-divider circuit, changes in FSR resistance are converted into corresponding voltage variations that can be measured using the analogue input of a microcontroller. Their thin and flexible construction allows them to be incorporated into beds, mattresses, pillows and other supporting surfaces for unobtrusive monitoring.

Lokavee et al. (2024) developed a sleep-monitoring system using FSR sensors embedded in a pillow sheet. The FSRs detected changes in pressure distribution caused by movements around the head and pillow area. The study also described the relationship between applied force, sensor resistance, conductance and output voltage. The relationship between the applied force and the electrical resistance and conductance of an FSR is illustrated in Figure 2.6. Although Lokavee et al. (2024) characterised an FSR 406 sensor, the figure demonstrates the general force-resistive operating principle, in which resistance decreases and conductance increases as the applied force increases. A fixed resistor and the variable FSR resistance were connected as a voltage-divider circuit, allowing force-related resistance changes to be

represented as measurable voltage signals. The researchers evaluated the system against polysomnography and demonstrated the potential of FSR sensors for unobtrusive bed-related monitoring. However, the study mainly concentrated on body movement, breathing activity and sleep-related breathing disorders rather than direct heart-rate estimation.

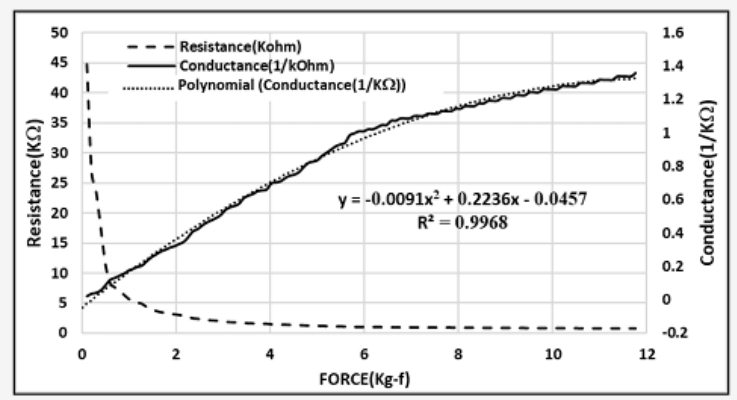


Figure 6: Relationship between applied force, resistance and conductance of an FSR (Lokavee et al., 2024)

A study more closely related to heart-rate monitoring was conducted by Haghi et al. (2023), who developed an unconstrained cardiorespiratory monitoring system using two FSR 408 strip sensors. The sensors were installed horizontally on a slatted bed base beneath the mattress. One sensor covered the thoracic region, while the other covered the abdominal region. This arrangement allowed the sensors to detect mechanical movements associated with cardiac and respiratory activity without requiring the participants to wear the FSR sensors. Figure 5 illustrates the under-mattress placement of the FSR 408 strip sensors used by Haghi et al. (2023).

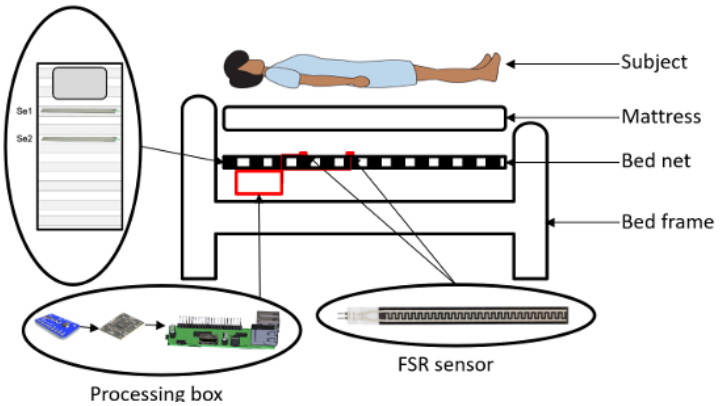


Figure 2.6 Under-mattress placement of FSR 408 strip sensors for cardiorespiratory monitoring (Haghi et al., 2023)

Haghi et al. (2023) evaluated the system using 20 healthy subjects consisting of 12 males and eight females. Each participant was monitored for 80 seconds in four sleeping positions which are prone, right lateral, supine and left lateral. The FSR signals were acquired at a sampling frequency of 150 Hz. Before processing, the first and last 10 seconds of each recording were removed to reduce transient disturbances associated with the beginning and end of the measurement. The researchers applied filtering and maximal overlap discrete wavelet transform multiresolution analysis to extract the cardiac component and estimate HR.

The FSR measurements obtained by Haghi et al. (2023) were evaluated against reference measurements. The system produced an overall mean absolute HR error of 3.24 bpm. The sensor covering the thoracic region generally demonstrated better performance than the abdominal sensor, indicating that sensor placement influences the quality of the detected cardiac signal. However, the performance also varied according to sleeping position and participant characteristics. These findings show that FSR 408 strip sensors can detect small cardiac-related mechanical movements through a mattress but that the resulting signal depends on the mechanical coupling between the subject, mattress and sensor.

The findings of Haghi et al. (2023) are particularly relevant to the present study because both systems use FSR 408 strip sensors for bed-based cardiac monitoring. Nevertheless, the present study differs by using three FSR 408 sensors positioned across the thoracic region and evaluating each sensor independently against a BITalino ECG reference. The use of three sensors allows the effect of sensor location and differences in pressure distribution to be investigated. Overall, FSRs provide a suitable sensing method for the proposed system because they can be connected through a relatively simple voltage-divider circuit and integrated into a thin sensor pad. However, signal processing and performance evaluation remain necessary because FSR measurements may be affected by sensor placement, body movement, mattress characteristics and variations among subjects.

2.5 SIGNAL PROCESSING FOR BED-BASED HEART-RATE MONITORING

The mechanical movements generated by cardiac activity are small compared with the pressure produced by a patient's body weight and major body movements. Consequently, raw signals obtained from bed-based sensors may contain baseline drift, respiratory components, electrical noise and motion artefacts in addition to the desired cardiac information. Signal-processing methods are therefore required to reduce unwanted components and identify cardiac-related peaks that can be used to estimate heart rate. Common processing stages include signal filtering, normalization, peak detection and comparison with a reference ECG signal.

2.5.1 Signal Filtering and Normalization

Filtering is applied to separate cardiac-related components from other frequencies contained in a raw bed-based signal. Low-frequency fluctuations may be caused by respiration, changes in body pressure and sensor drift, whereas higher-frequency components may

originate from electrical interference, sudden movement and measurement noise. A suitable bandpass filter can retain the frequency components associated with cardiac activity while attenuating unwanted low- and high-frequency components.

Different filtering methods and frequency ranges have been employed depending on the sensor and monitoring system. Xu et al. (2020) initially applied a zero-phase low-pass filter with a cutoff frequency of 30 Hz and a 50 Hz notch filter to reduce high-frequency and power-line interference in a PVDF mattress signal. A finite impulse response bandpass filter between 4 and 16 Hz was subsequently used to extract the BCG component for HR estimation. In comparison, Haghi et al. (2023) used a Chebyshev Type I bandpass filter between 2.5 and 5 Hz before applying wavelet-based processing to the cardiac component obtained from FSR 408 sensors. These differences demonstrate that the selected filtering range depends on the sensing technology, signal characteristics and processing algorithm. The signal processing stages employed by Haghi et al. (2023) to extract cardiac and respiratory components from the FSR signal are illustrated in Figure 2.7.

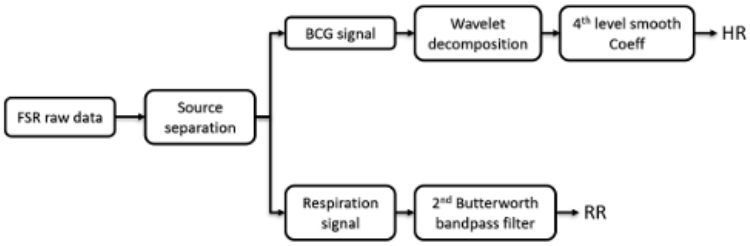


Figure 2.7 Signal processing procedure for physiological signals obtained from FSR 408 sensors (Haghi et al., 2023)

Haghi et al. (2023) also removed the first and last 10 seconds from each 80-second recording before processing. Removing these portions reduced transient disturbances that could occur while the subject settled into position or prepared to end the measurement. This

procedure allowed the middle section of the recording, in which the subject was more stable, to be used for analysis.

After filtering, normalisation can be applied to place signals with different amplitude ranges on a common scale. This is particularly useful when comparing signals obtained from multiple FSR sensors because differences in body pressure, sensor position and mechanical coupling may cause each sensor to produce a different voltage range. Min-Max normalisation transforms the signal values into a specified range, commonly between zero and one, while maintaining the relative waveform pattern. Normalisation does not remove noise or improve the original signal quality, but it facilitates waveform comparison and the use of consistent peak-detection thresholds across different sensor channels.

2.5.2 Peak Detection and ECG Validation

After filtering and normalisation, cardiac-related peaks can be detected from the processed signal. In ECG, the R-peak is normally used to identify individual cardiac cycles because it is the most prominent component of the QRS complex. In a BCG or pressure-based signal, a mechanical cardiac peak occurs after the ECG R-peak because the mechanical response follows the electrical activation of the heart, as illustrated by Pino et al. (2017) in Figure 2.1. The time intervals between consecutive detected cardiac peaks can then be used to estimate HR. An example of the cardiac component extracted from an FSR signal and the corresponding peak-detection result is presented in Figure 2.8 (Haghi et al., 2023).

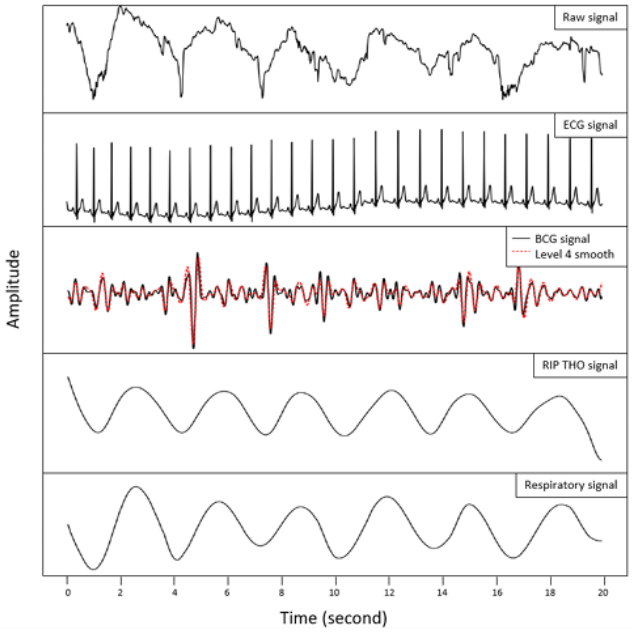


Figure 2.8 Example of cardiac-signal extraction and peak detection from an FSR signal (Haghi et al., 2023)

Peak-detection parameters must be selected carefully because noise and secondary waveform components may cause multiple peaks to be detected within one cardiac cycle. Xu et al. (2020) applied a minimum separation related to 0.40 seconds between adjacent BCG J-waves during HR estimation. The use of a minimum peak distance can reduce repeated detections within a single heartbeat. A peak-height threshold may also be applied to prevent small signal fluctuations from being incorrectly classified as cardiac peaks. However, thresholds that are too high may exclude valid low-amplitude peaks, while thresholds that are too low may increase false detections.

ECG is commonly used as a reference when validating heart-rate measurements obtained from bed-based mechanical sensors. Haghi et al. (2023) compared the HR estimated from FSR 408 sensors with reference cardiac measurements and evaluated the performance using absolute error and agreement analysis. The study obtained an overall mean absolute HR error

of 3.24 bpm, demonstrating that cardiac activity detected through an FSR sensor could be evaluated against a standard reference.

For ECG–FSR comparison, an FSR peak occurring after an ECG R-peak can be considered a corresponding mechanical cardiac event when it falls within an appropriate time interval. A one-to-one matching procedure can be used to prevent one FSR peak from being assigned to multiple ECG R-peaks. The number of successfully matched peaks indicates the ability of the FSR sensor to detect corresponding cardiac events, while the variation in the ECG-to-FSR delay indicates the consistency of the temporal relationship between the electrical and mechanical signals. Therefore, HR error, matched-beat percentage and ECG-to-FSR delay variability can be used together to provide a broader evaluation of a bed-based sensor than HR estimation alone.

2.6 COMPARISON OF PREVIOUS STUDIES

Previous studies have employed different sensing technologies, sensor placements and signal-processing methods for bed-based physiological monitoring. Fibre-optic sensors offer high sensitivity and immunity to electromagnetic interference but require specialised optical equipment. Piezoelectric sensors can detect small mechanical vibrations through a mattress, although they require amplification and signal-conditioning circuits. Load cells provide unobtrusive monitoring from beneath the bed legs but may be affected by changes in whole-body position and load distribution. In comparison, FSR sensors offer a relatively simple and flexible sensing method that can be integrated into a mattress, pillow or sensor pad.

Table 2.1 summarises the sensing technology, sensor placement, measured parameters, sampling rate and principal findings of the previous bed-based monitoring studies.

Table 2.1 Comparison of previous bed-based monitoring studies

Study	Sensor and placement	Parameters	Sampling rate	Main finding/limitation
Haghi et al. (2023)	Two FSR 408 strips beneath the thoracic and abdominal regions	HR and RR	150 Hz	Achieved a mean HR error of 3.24 bpm; the thoracic sensor generally performed better.
Lokavee et al. (2024)	FSR sensors embedded in a pillow sheet	Sleep and respiratory activity	Not reported	Demonstrated simple FSR integration but did not focus on HR estimation.
Nishimura et al. (2025)	Four load sensors beneath the bed leg	RR	10 Hz	Provided unobtrusive monitoring but required a sensor beneath each bed leg.
Xu et al. (2020)	PVDF sensor integrated into a mattress	HR and RR	100 Hz	Detected cardiac activity through the mattress but was affected by body movement.
Zhu et al. (2015)	FBG sensor arrays beneath a mattress	RR	128 Hz	Sensitive and unaffected by electromagnetic interference, but requires costly optical equipment.

2.7 CHAPTER SUMMARY

This chapter reviewed the concepts and previous studies related to non-invasive bed-based heart-rate monitoring. Conventional ECG provides reliable cardiac measurements but requires electrodes to be attached to the patient. In contrast, ballistocardiography detects cardiac activity indirectly through mechanical movements transmitted through the body and bed.

Several sensing technologies were reviewed, including fibre-optic, piezoelectric, load and FSR sensors. The FSR 408 was selected because of its thin and flexible structure, large sensing area, simple voltage-divider circuit and low implementation cost. However, its performance may be affected by sensor placement, body movement, mattress properties, pressure distribution and subject differences.

Signal-processing methods such as filtering, normalisation, peak detection and ECG validation were also discussed. Limited research has independently compared multiple FSR 408 sensors across the thoracic region using heart-rate error, matched-beat percentage and ECG-to-FSR timing variability. Therefore, this study addresses this gap by evaluating a three-sensor FSR pad against a BITalino ECG reference for heart-rate monitoring in bedridden patients.

CHAPTER 3

METHODOLOGY

3.1 INTRODUCTION

This chapter presents the methodology used to design, develop and evaluate the proposed non-invasive heart-rate monitoring system for bedridden patients. The system consists of three FSR 408 strip sensors arranged in a sensor pad and positioned on top of the mattress beneath the subject at the thoracic region to detect small pressure variations produced by cardiac activity. The sensors were connected to an Arduino Uno through voltage-divider circuits for data acquisition, while a BITalino ECG served as the reference measurement. The acquired FSR and ECG signals were processed in MATLAB through signal synchronisation, data trimming, bandpass filtering, Min-Max normalisation and peak detection. The detected ECG R-peaks were subsequently matched with the corresponding FSR peaks to estimate heart rate and evaluate each sensor using HR error, matched-beat percentage and ECG-to-FSR timing variability.

3.2 SYSTEM DESCRIPTION

3.2.1 System Process Flow

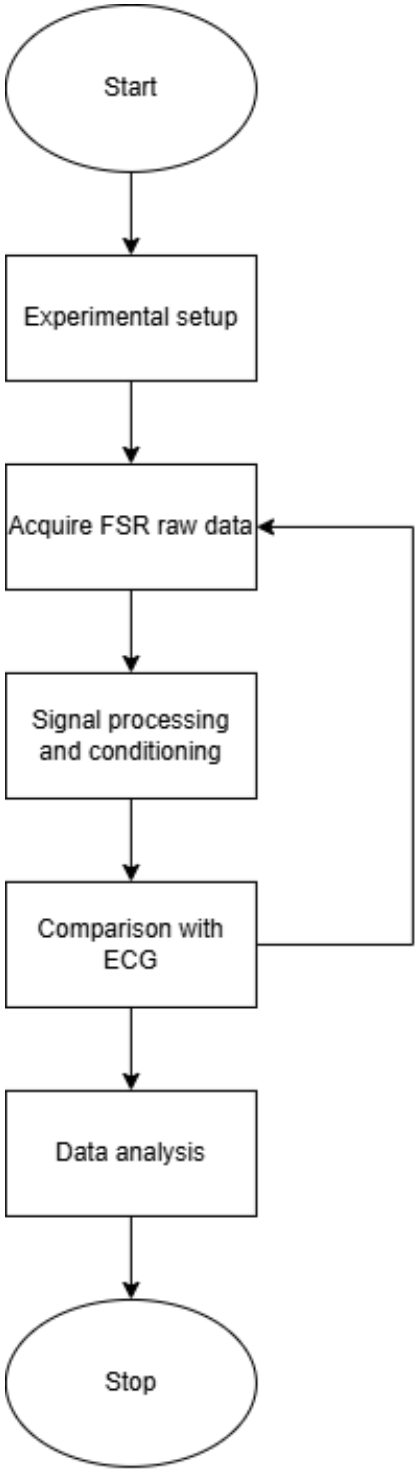


Figure 3.1 Process Flow of the System

The methodology follows a structured, step-by-step operational sequence designed to ensure the development of a cost-effective, unconstrained health monitoring system that integrates seamlessly into standard furniture. As outlined in the system architecture, the process initiates with the experimental setup where the sensors are positioned a mattress to monitor patients in supine position. From there, raw FSR data acquired will capture pressure and voltage changes continuously as subjects lie on the bed.

Once the raw data are gathered, they undergo signal processing and conditioning which involves filtering, amplification and analog-to-digital conversion to eliminate transient noise and separate the raw signals into distinct cardiorespiratory components like the BCG. Following conditioning, the extracted heart rate undergoes a comparison with ECG reference signals to validate their reliability and ensure clinical accuracy.

Finally, the workflow concludes with a comprehensive data analysis to evaluate the performance of the FSR sensors against the ECG reference. The evaluation includes the ECG and FSR heart rates, absolute heart rate error, matched-beat percentage, mean ECG-to-FSR delay and delay standard deviation. These measurements are used to determine whether the FSR sensor system can detect cardiac activity consistently and provide heart rate readings comparable to the reference ECG, thereby assessing its potential as an unobtrusive and cost-effective system for long-term health monitoring.

3.2.2 Experimental Setup

The experimental setup is designed to evaluate and validate the unconstrained heart rate monitoring system using accessible and cost-effective components integrated with microcontroller technology. Multiple FSR sensors were placed on the bed which is in thoracic region. To ensure precise anatomical alignment, the sensor pad placement is guided by anthropometric measurements of the human body. The sensor pad was positioned approximately at 72% of the subject's total body height measured from the heel which is corresponding to the targeted thoracic region. The anthoropometric reference used for determining the sensor placement is shown in Figure 3.2 and the overall experimental setup is illustrated in Figure 3.3 where it showing the placement of the sensor pad on the bed and its conection to the electronic component.

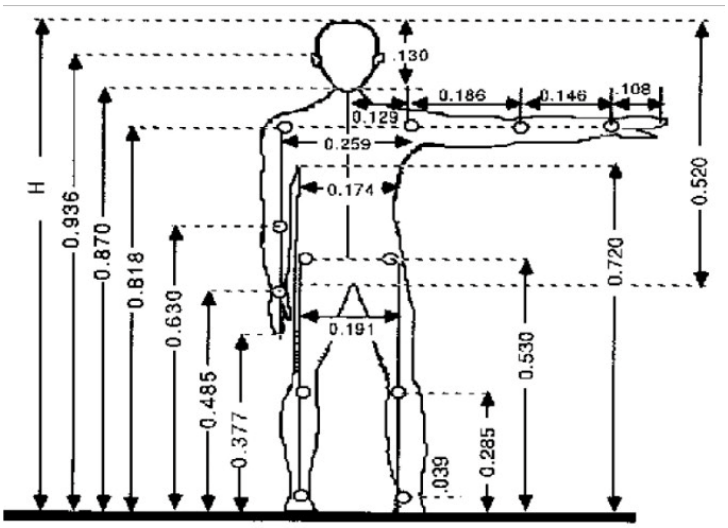


Figure 3.2 Anthropometry of the Human Body

Table 3.1 summarises the height, weight, and calculated sensor placement for the four subjects. Although all subjects maintained the same supine position, the position of the sensor array was adjusted according to each subject’s height to align it with the thoracic region. The placement distance was calculated as 0.72 times the subject’s height and measured from the

heels. Differences in body dimensions and weight may have influenced the pressure transferred to the FSR sensor array and, consequently, the amplitude and quality of the recorded signals.

Table 3.1 Subject characteristics and calculated FSR sensor placement

Subject	Height (cm)	Weight (kg)	Calculated placement from heels (cm)
1	165	60	118.8
2	161	47	115.9
3	157	82	113.0
4	156	45	112.3

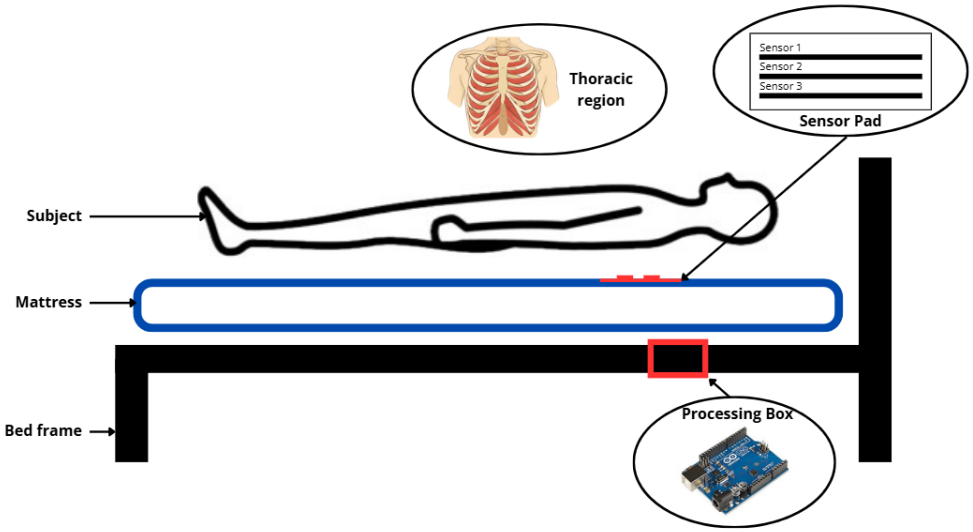


Figure 3.3 Experimental Setup

The proposed circuit was designed and tested using Tinkerad prior to physical assembly. The simulation was conducted to verify the circuit configuration and ensure that the FSR sensors could be appropriately interfaced with the Arduino Uno microcontroller before proceeding. The overall circuit simulation is presented in Figure 3.4 and Figure 3.5 respectively.

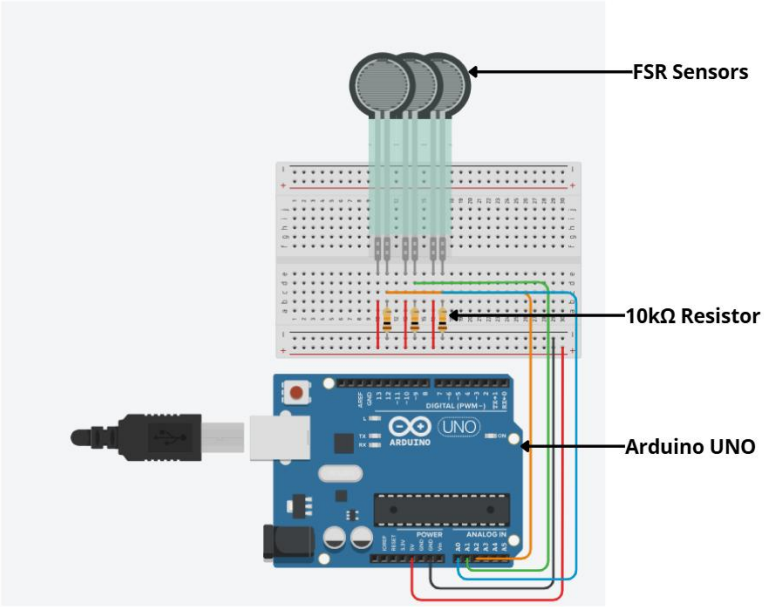


Figure 3.4 Circuit Simulation in Tinkercad

In the Tinkercad simulation, circular FSR 402 sensors were used because they were the only FSR model available on the platform instead of the strip-shaped FSR 408 sensors. The FSR 402 was used only to verify the voltage-divider circuit and Arduino analog-input connections because both sensor models operate according to the same force-resistive principle, in which the sensor resistance decreases as the applied force increases. The physical system was subsequently implemented using three FSR 408 strip sensors to provide a larger sensing area across the thoracic region

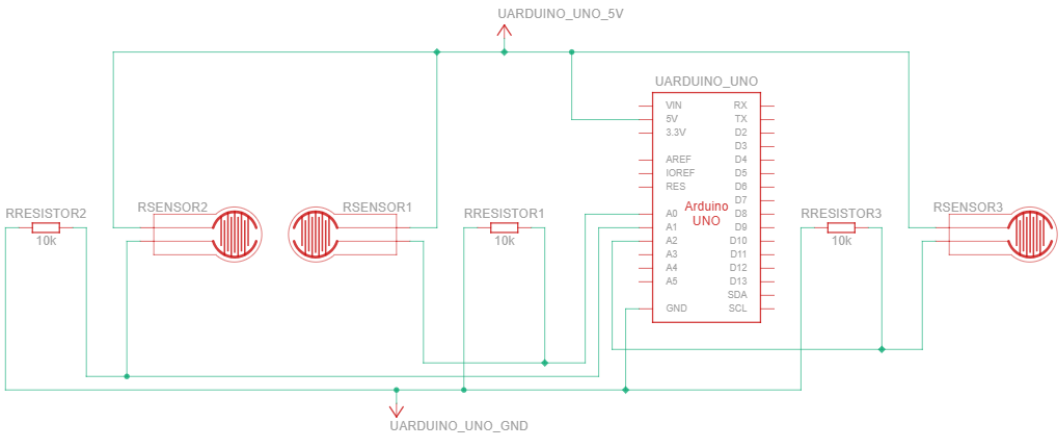


Figure 3.5 Schematic Diagram of the Circuit

FSR 408 model was selected for this experiment due to its thin and flexible structure for a seamless and unobstructive contact for comfortable experience user while detecting the subtle pressure variations caused by cardiac activity. To properly interface the analog FSR sensor with the Arduino Uno, a voltage divider circuit is constructed with a 10k Ohm resistor as shown in Figure 3.6. It allowed changes in pressure to be translated into corresponding voltage fluctuations that could be acquired through the microcontroller's analog input. After successful verification through simulation, the circuit was physically assembled and tested using the same electrical configuration as shown in Figure 3.7.

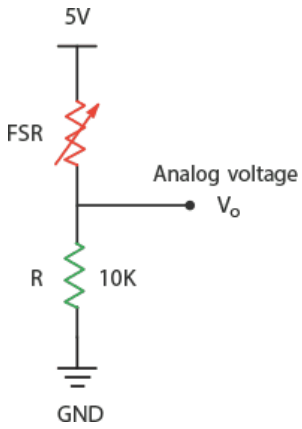


Figure 3.6 Voltage Divider

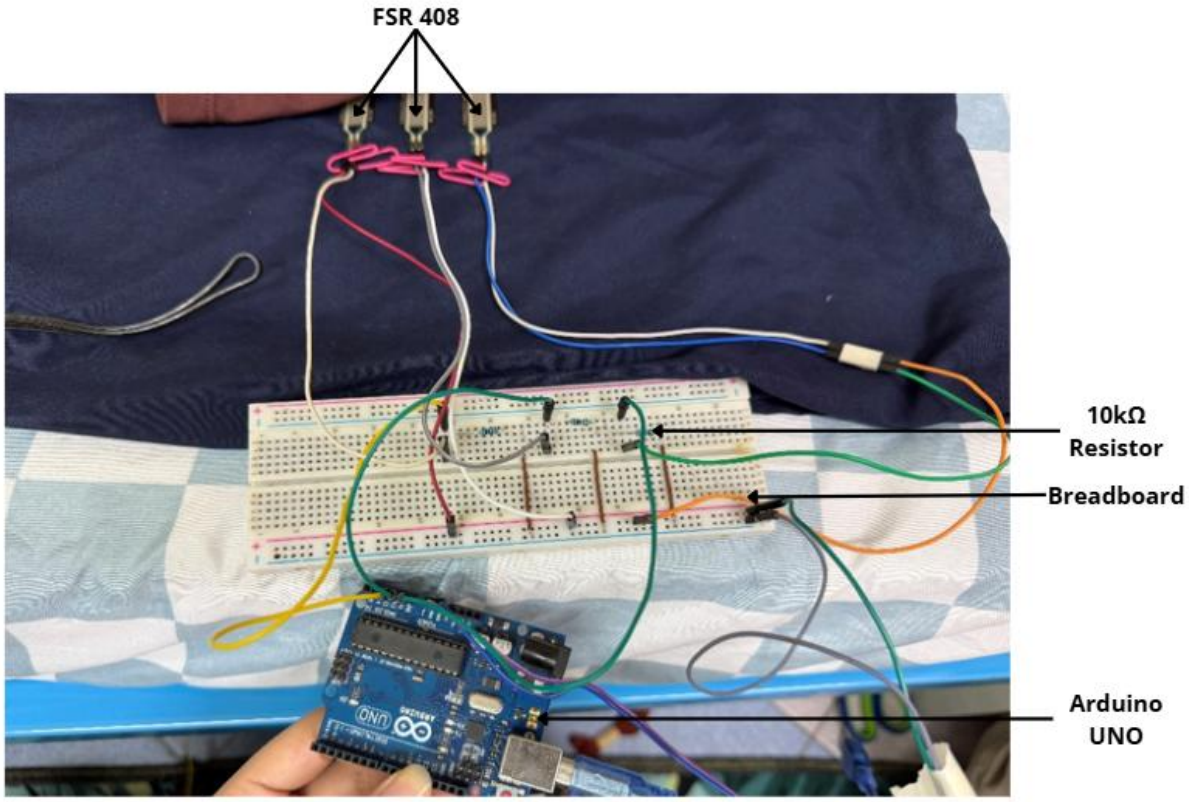


Figure 3.7 Physical Implementation of the FSR Sensor Circuit

To provide pressure distribution and sensing coverage across the testing area, multiple FSR strips are arranged in a multi-channel configuration. The sensors were position horizontally to the bed with a 1cm spacing between adjacent sensors. This arrangement was intended to provide distributed sensing coverage across the designated measurement area. This is also to investigate variations in pressure signals detected at different positions on the chest wall. The sensor array arrangement and the inter-sensor spacing are illustrated in Figure 3.8.

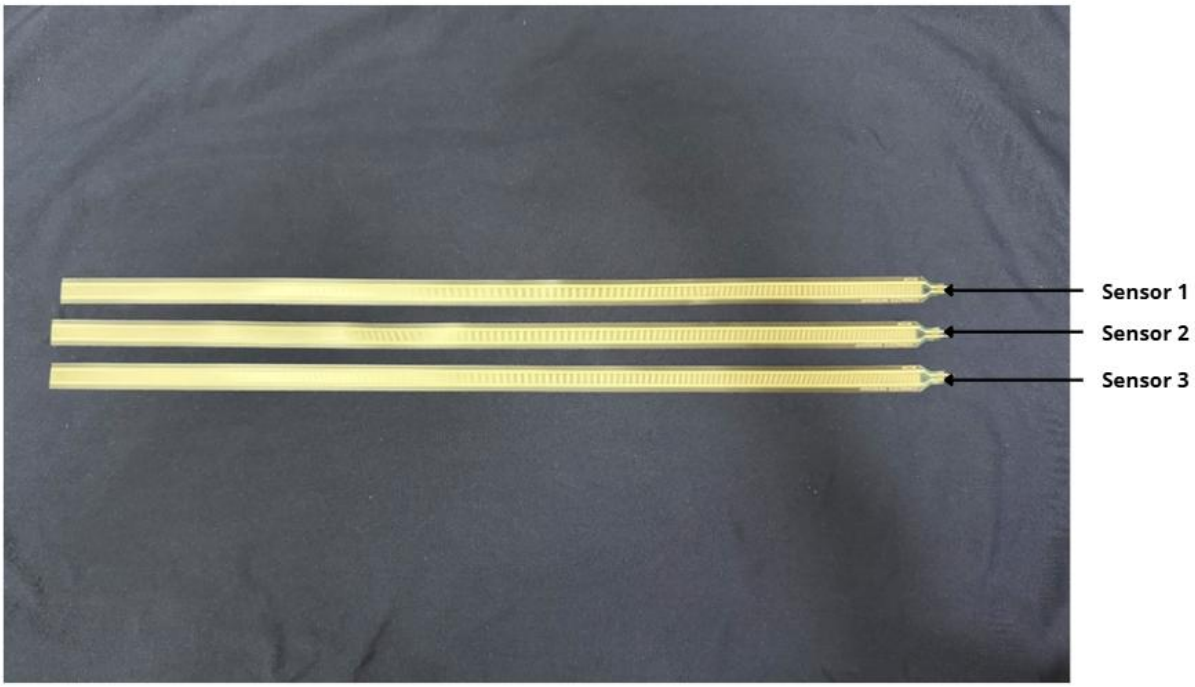


Figure 3.8 Sensor Array Layout Showing 1cm Spacing

The experimental trials focus exclusively on the supine position, where the subject lies flat on their back. This position was selected to ensure consistent contact between the subject's body and the sensor. To determine the appropriate placement of the sensor pad, the subject's body length was measured from the heel to the targeted body part to placed the Sensor 2 using a measuring tape. Sensor 2 was approximately to be placed 72% of the measured body length from the heel. This position was selected to locate the sensor closer to the cardiac region and to provide a consistent reference point for sensor placement across the experimental trials. The measurement procedure and sensor pad placement are illustrated in Figure 3.9.

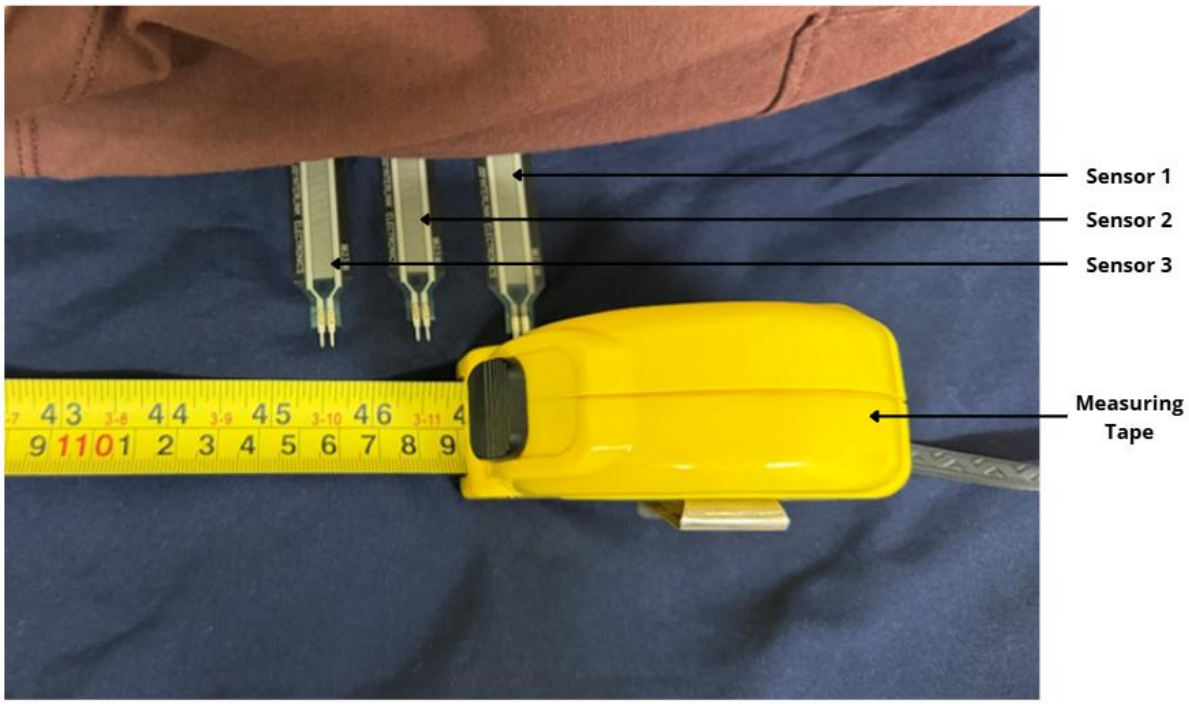


Figure 3.9 Subject Positioning, Body-Length Measurement and Sensor Placement

3.2.3 Data Acquisition And Signal Processing

The data acquisition and signal processing workflow was developed to convert raw physical pressure changes into reliable cardiac timing information for heart rate. The FSR sensors were interfaced with an Arduino Uno microcontroller to continuously acquire analog voltage signals corresponding to changes in pressure applied to the thoracic sensor array. The FSR sensors are sampled at a frequency of 100 Hz using a non-blocking millis() timing loop to capture dynamic resistance changes corresponding to cardiac activity.

The acquired FSR data were subsequently transferred to MATLAB for signal processing. A bandpass filter with passband frequencies of 0.5–5 Hz was applied to the raw FSR signals to reduce unwanted signal components. The lower cutoff frequency of 0.5 Hz was selected to reduce low-frequency baseline fluctuations, including those associated with respiration and sensor drift, while the upper cutoff frequency of 5 Hz was used to attenuate

higher-frequency noise. Following filtering, Min-Max normalization was applied to scale each FSR waveform to a range of 0 to 1. The processed signals were then subjected to automated peak detection using MATLAB's findpeaks function to identify the systolic pulse peaks. The detected peaks were subsequently used for temporal comparison with the reference ECG signal and Pulse Transit Time (PTT) estimation.

The ECG signal was acquired to be the gold standard references. ECG signal from BITalino biosignal acquisition device was connected via Bluetooth and monitored in real time using the OpenSignals companion software interface at a sampling frequency of 100Hz. A three-lead ECG configuration was used to capture the electrical activity of the heart. The positive (Red) electrode was placed under the left collarbone (clavicle), the negative (Black) electrode was positioned on the right clavicle and the reference (White) electrode was attached to a low-muscle bony landmark where in this experiment the elbow was selected.

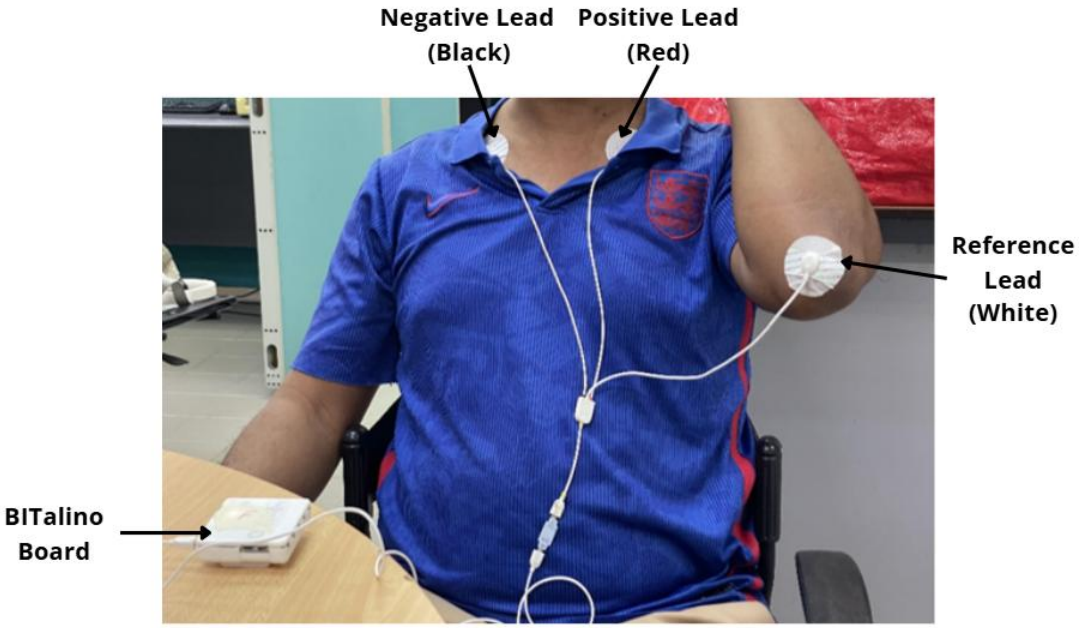


Figure 3.10 ECG Acquisition Setup

The BITalino hardware records the ECG signal as raw analog-to-digital converter (ADC) values. These values were converted into physical voltage units using the device's transfer function. The ECG voltage in volts was calculated using the following equation:

$$ECG(V) = \frac{(\frac{ADC}{2^n} - \frac{1}{2}) \times VCC}{G_{ECG}} \quad (1)$$

$$ECG(mV) = ECG(V) \times 1000 \quad (2)$$

$VCC = 3.3V$ (operating voltage)

$G_{ECG} = 1100$ (sensor gain)

$ECG(V)$ – ECG value in Volt (V)

$ECG(mV)$ – ECG value in milliVolt (mV)

ADC – Value sampled form the channel

n – Number of bits of the channel

The converted ECG signal was used as the reference signal for identifying the electrical cardiac events and corresponding cardiac cycles. While your thoracic FSR sensor measures mechanical pressure variations and were represented either as raw ADC values or normalized values ranging from 0 to 1. Since both systems operated at a sampling frequency of 100 Hz, each sample corresponded to a temporal resolution of 10 ms. Therefore, the processed ECG and FSR signals were represented on the same sampling time grid enabling the temporal relationship between the electrical cardiac activity and the mechanically induced pressure waveform to be investigated.

This common sampling frequency enabled the temporal relationship between the electrical cardiac activity detected by the ECG and the mechanical pulse waveform detected by the FSR sensors to be investigated. In the ECG signal, ventricular depolarization produces

a prominent R-peak, which was used as the reference point for each cardiac cycle. The corresponding mechanical response was observed as a systolic pulse peak in the FSR waveform, occurring after the ECG R-peak due to the time required for the pressure wave generated by cardiac contraction to reach the measurement site. This is because electrical signals propagate through the body virtually instantly while blood flow takes a fraction of a second to travel through the arterial tree, a distinct temporal offset occurs between the electrical and mechanical events.

The FSR and ECG signals were initially recorded for approximately 80 seconds per trial. Prior to filtering and signal normalization, the recordings were synchronized and trimmed to reduce the influence of transient effects at the beginning and end of the recording. The first 10 seconds and the final 10 seconds of each recording were removed, resulting in a 60 seconds segment used for subsequent analysis. The ECG and FSR datasets were then adjusted to contain the same number of samples to ensure that both signals shared a common temporal reference.

To quantify the temporal delay between these electrical and mechanical events, Pulse Transit Time (PTT) was calculated. PTT represents the time interval between the occurrence of an ECG R-peak and the corresponding systolic peak detected by the FSR sensor. It was calculated using:

$$PTT = t_{FSR} - t_{ECG} \tag{3}$$

$t_{\text{ECG_peak}}$ = the time at which the ECG R-peak occurs

$t_{\text{FSR_peak}}$ = the time at which the corresponding FSR systolic peak occurs

In MATLAB, the detected ECG R-peaks were used as reference points for identifying the corresponding FSR systolic peaks within each cycle. For each ECG R-peak, the algorithm

identified all eligible FSR peaks occurring within a predefined ECG to FSR mechanical delay window and selected the peak with the highest normalized amplitude. A one to one matching procedure was applied to ensure that each FSR peak could only be associated with one ECG R-peak. A minimum delay of 0.05 s (50 ms) and a maximum delay of 0.40 s (400 ms) were used as a broad matching window. This range was selected based on previous ballistocardiographic studies showing that the mechanical cardiac response occurs after the ECG R-wave, with reported R-to-J timing commonly occurring within several hundred milliseconds. Gómez-Clapers et al. (2014) considered a 400 ms window following the ECG R-peak for identifying the BCG J-peak. The lower limit of 50 ms was included to prevent immediate ECG-related noise or non-physiological peaks from being incorrectly matched. The valid PTT values were subsequently converted from seconds to milliseconds (ms) and used for statistical analysis and comparison across the experimental trials and sensor locations.

3.2.4 Data Analysis

To ensure data reliability and minimize the influence of transient disturbances, each experimental session is conducted for approximately 80 seconds per trial and repeated for a total of three trials. The recorded ECG and FSR signals are first synchronized according to their common recording duration. The first 10 seconds and the last 10 seconds of the synchronized data are then removed to minimize the effects of initial sensor adjustment, subject movement, and possible signal instability at the beginning and end of the recording. Therefore, approximately 60 seconds of data from each trial are used for the subsequent analysis.

The data analysis evaluates the accuracy and reliability of the developed unconstrained heart rate monitoring system by comparing the heart rate obtained from the FSR sensor pad with the reference ECG signal. The ECG is used as the gold standard reference because it

directly provides the electrical cardiac activity from which the R-peaks and corresponding heart rate are determined. The FSR signals are treated as the measured output of the proposed sensor pad and are evaluated against the ECG reference.

To determine whether an FSR pulse corresponds to an ECG cardiac event, each detected ECG R-peak is compared with subsequent FSR pulse peaks within a predefined physiological time window of 50–400 ms. This window represents the expected temporal delay between the electrical cardiac event detected by the ECG and the subsequent mechanical pulse detected by the FSR sensor. A one-to-one matching procedure is applied such that each detected FSR pulse can only be matched to one ECG R-peak. This prevents a single FSR pulse from being counted multiple times and provides a more reliable assessment of the FSR sensor's beat detection capability. The number of successfully matched beats is then used to determine the matched-beat percentage, with the ECG-detected beats serving as the reference:

$$Matched\ Beat\ Percentage = \frac{N_{matched}}{N_{ECG}} \times 100\% \quad (4)$$

N_{matched} = number of FSR beats successfully matched to ECG R-peaks

N_{ECG} = total number of beats detected from the ECG reference

The temporal relationship between the ECG and FSR signals is also evaluated by calculating the delay between each matched ECG R-peak and FSR pulse peak:

$$Delay = t_{FSR} - t_{ECG} \quad (5)$$

$$\bar{D} = \frac{1}{N}\sum_{i=1}^{N} D_i \quad (6)$$

t_{FSR} = the time of the detected FSR pulse peak

t_{ECG} = time of the corresponding ECG R-peak

D_{i} =. the ECG-to-FSR delay for each matched beat

N = the number of matched beats

The resulting delay values are expressed in milliseconds. The mean delay and standard deviation are calculated for each FSR channel to evaluate the consistency of the detected mechanical pulse relative to the ECG reference.

Heart rate was calculated independently from the ECG and each FSR channel using the time intervals between consecutive detected peaks. For the ECG signal, the intervals between consecutive R-peaks were used, while the intervals between consecutive independently detected FSR peaks were used for the FSR channels. Heart rate was calculated using:

$$HR = \frac{60}{\bar{T}} \quad (7)$$

HR = the estimated heart rate in beats per minute (bpm)

$\bar{T}$ = the mean interval between consecutive detected peaks in seconds

The agreement between the FSR-derived heart rate and the ECG reference heart rate was evaluated using absolute heart rate error. The absolute heart rate error was calculated separately for each FSR channel using:

$$Absolute\ HR\ Error = |HR_{FSR} - HR_{ECG}| \quad (8)$$

HR_{FSR} = the heart rate calculated independently from the FSR peaks

HR_{ECG} = the heart rate calculated from the ECG R-peaks.

A smaller absolute heart rate error indicates closer agreement between the FSR measurement and the ECG gold-standard reference. The percentage heart rate error was also calculated using:

$$HR\ Error\ (\%) = \frac{|HR_{FSR} - HR_{ECG}|}{HR_{ECG}} \times 100 \quad (9)$$

The performance of all three FSR channels is evaluated independently for each of the three experimental trials. The results from the repeated trials are then compared to determine the consistency and reliability of the sensor pad under the same experimental conditions. The overall performance is assessed using the matched-beat percentage, mean ECG-to-FSR pulse delay and standard deviation of the delay.

Based on these performance measures, the suitability of the FSR sensor pad for unconstrained heart rate monitoring is determined. A high matched-beat percentage and consistent ECG-to-FSR pulse delay indicate that the FSR sensor is capable of reliably detecting cardiac activity and producing heart rate measurements that agree with the ECG gold-standard reference.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 INTRODUCTION

This chapter presents and discusses the experimental results obtained from the developed bed-based cardiac monitoring system using three FSR sensors. Data were collected from four subjects while maintaining the same supine body position. The signals recorded by the FSR sensor array were compared with electrocardiogram (ECG) signals acquired simultaneously using the BITalino device which served as the reference measurement.

The raw ECG and FSR signals were processed in MATLAB through signal synchronization, trimming, bandpass filtering and Min–Max normalization. Peak-detection algorithms were then applied to identify the ECG R-peaks and the corresponding cardiac-related peaks in each FSR signal. The detected peaks were used to estimate heart rate and calculate pulse transit time (PTT) between the ECG and FSR signals.

The results are presented through signal plots, ECG–FSR waveform comparisons, heart rate measurements, matched-beat counts and PTT values. The performance of the three FSR sensors is evaluated across all four subjects to determine which sensor provides the most consistent cardiac signal and the closest agreement with the ECG reference. Variations between the sensors and subjects are also discussed in relation to sensor placement, body-pressure distribution, and the transmission of cardiac-induced mechanical vibrations through the mattress.

4.2 RESULTS AND ANALYSIS

4.2.1 Raw And Processed Signal Results

The raw signals acquired from the BITalino ECG and the three FSR sensors for the four subjects were analysed to examine the quality of the recorded cardiac activity. Each recording was conducted for approximately 80 seconds, with all subjects maintaining the same supine position to minimise variations caused by changes in body posture. Nevertheless, differences in signal amplitude and waveform characteristics were expected because of subject-specific factors such as body structure, body mass, thoracic position, and pressure distribution over the sensor array.

Before signal processing, the ECG and FSR recordings were synchronised based on the shorter recording length. The first and last 10 seconds were subsequently removed to minimise transient disturbances associated with the subject settling onto the bed at the beginning and moving at the end of the recording. Therefore, approximately 60 seconds from the middle of each recording were retained for analysis.

Figure 4.1 presents the trimmed raw ECG and FSR signals obtained from Subject 1. The raw ECG signal contained identifiable cardiac activity, while the raw FSR signals represented the mechanical pressure changes transmitted from the subject's body through the mattress. Compared with the ECG signal, the FSR signals exhibited greater low-frequency baseline fluctuations and variations in the pressure applied to each sensor. The three FSR channels also produced different signal amplitudes despite being recorded simultaneously. This variation may be attributed to the slightly different positions of the sensors beneath the thoracic region, which caused each sensor to receive a different amount of body pressure.

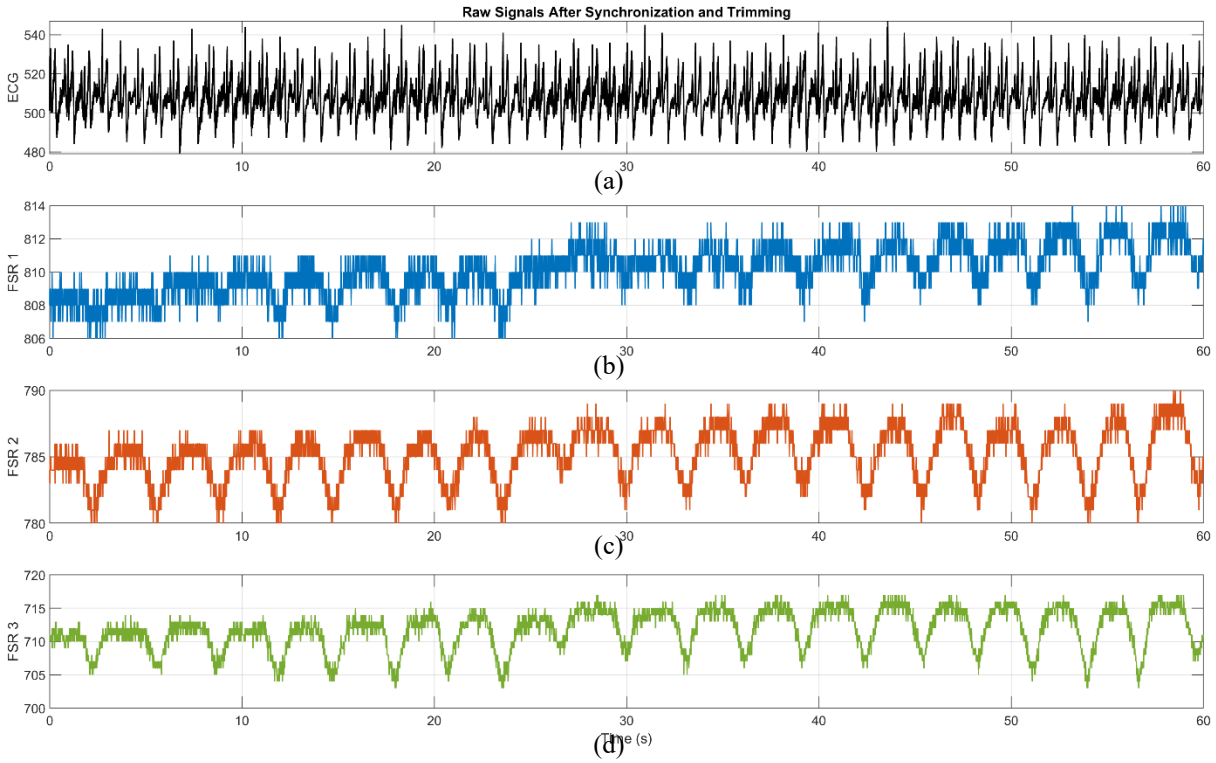


Figure 4.1 Trimmed raw ECG and FSR signals for Subject 1

(a) The raw ECG; (b) Sensor 1; (c) Sensor 2; (d) Sensor 3

Following synchronisation and trimming, the signals were processed in MATLAB. A bandpass filter with a frequency range of 0.5–5 Hz was applied to the FSR signals to reduce low-frequency baseline fluctuations and higher-frequency noise while retaining cardiac-related mechanical components. Meanwhile, the ECG signal was filtered within a frequency range of 0.5–40 Hz to reduce unwanted signal components while preserving the ECG waveform and its R-peaks.

Min–Max normalisation was then applied independently to each signal to scale its amplitude between 0 and 1. Normalisation allowed the ECG and FSR signals to be displayed and compared on a common scale despite their different original amplitude ranges and measurement characteristics. The normalisation process adjusted only the signal amplitudes and did not alter the timing of the detected cardiac events.

Figure 4.2 shows the filtered and normalised ECG and FSR signals obtained from Subject 1. Compared with the trimmed raw signals in Figure 4.1, the processed signals exhibited reduced baseline variations and more distinguishable cardiac-related peaks. The improvement in signal quality supported the subsequent processes of peak detection, heart rate estimation, and ECG–FSR peak matching.

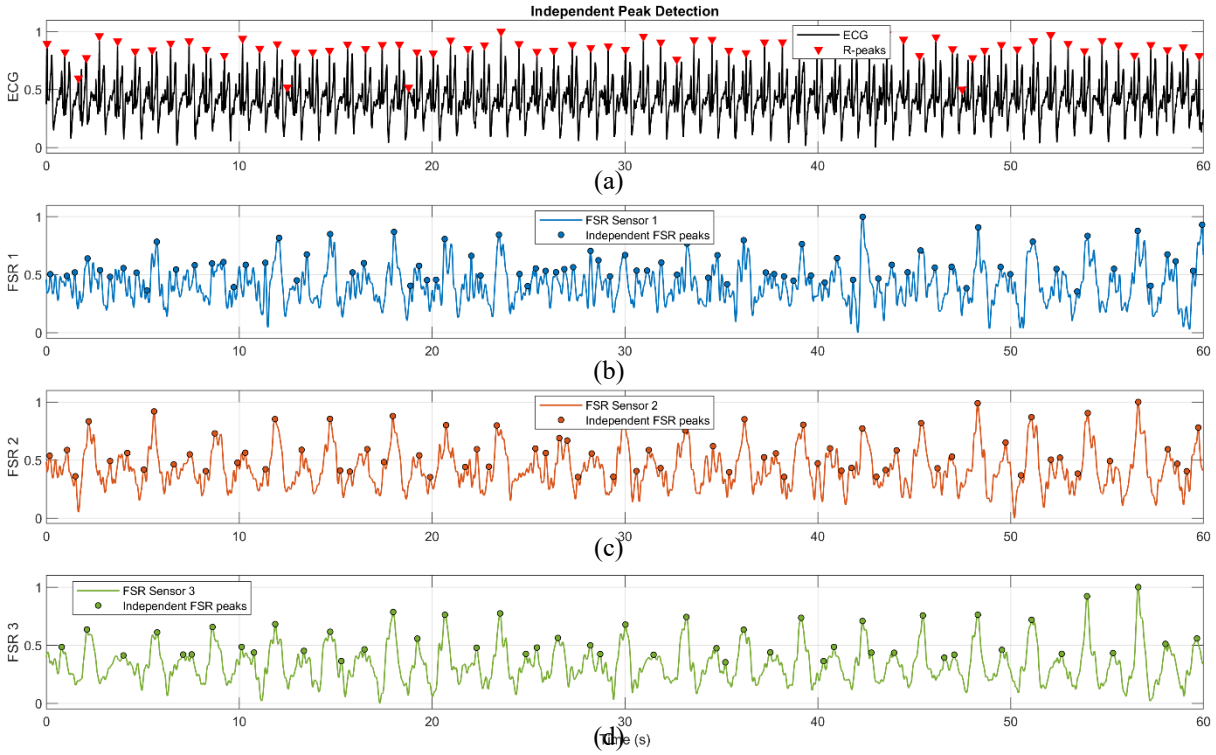


Figure 4.2 Filtered and normalised ECG and FSR signals for Subject 1

(a) The raw ECG; (b) Sensor 1; (c) Sensor 2; (d) Sensor 3

The same synchronisation, trimming, filtering, and normalisation procedures were applied to the recordings from Subjects 2, 3, and 4. Although all four subjects were tested under the same experimental conditions and in the same supine position, variations were observed among their FSR waveforms. These variations suggest that the mechanical cardiac signals detected on the mattress were affected not only by body position but also by subject-specific

physical characteristics and the amount of pressure transferred to each sensor. Therefore, all three FSR channels were retained for further analysis instead of relying on a single fixed sensor.

Subject 1 was selected as a representative example to demonstrate the effects of trimming, filtering, and normalisation. The same signal-processing procedure was applied to Subjects 2, 3, and 4, and their corresponding raw and processed signal plots are provided in Appendix A. Nevertheless, the quantitative results from all four subjects were included in the subsequent heart rate, PTT, and sensor-performance analyses.

4.2.2 Ecg And Fsr Peak-Detection Results

Following filtering and normalization, peak detection was performed on the ECG and three FSR signals. The ECG R-peaks were used as the reference cardiac events, while the peaks detected from the FSR signals represented the corresponding mechanical cardiac responses transmitted through the subject's body and mattress. Peak detection was performed independently for each signal because the amplitude and waveform characteristics differed among the subjects and sensor channels.

The selected recordings were Trial 4 for Subject 1, Trial 4 for Subject 2, Trial 3 for Subject 3, and Trial 3 for Subject 4. These recordings were selected to represent the peak-detection results because they provided suitable ECG–FSR heart rate agreement and overall sensor performance. Table 4.1 summarises the number of ECG and FSR peaks detected from the selected recording for each subject.

Table 4.1: Number of ECG and FSR peaks detected from the selected recordings

Subject	Selected trial	ECG peaks	FSR Sensor 1 peaks	FSR Sensor 2 peaks	FSR Sensor 3 peaks
1	4	72	83	74	49
2	4	89	88	45	55
3	3	66	51	46	54
4	3	86	90	49	83

As presented in Table 4.1, the number of FSR peaks varied across the subjects and sensor channels. For Subject 1, 72 ECG R-peaks were detected, while Sensors 1, 2, and 3 detected 83, 74, and 49 peaks, respectively. Sensor 2 produced a total peak count closest to the ECG reference. However, the similarity between the total peak counts did not necessarily indicate that all detected FSR peaks corresponded to the correct ECG R-peaks.

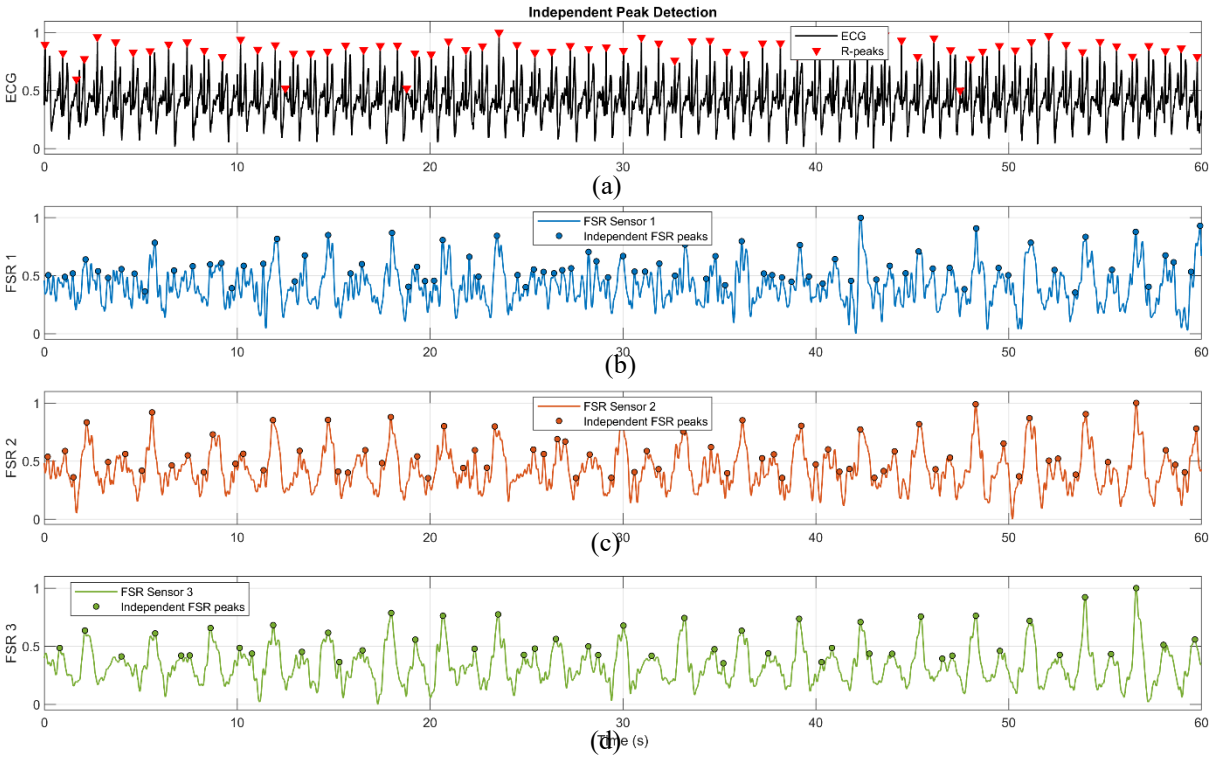


Figure 4.3 Detected ECG and FSR peaks for Subject 1, Trial 4

(a) The raw ECG; (b) Sensor 1; (c) Sensor 2; (d) Sensor 3

Figure 4.3 shows the detected peaks for Subject 1. Differences can be observed in the number and distribution of the detected peaks among the three FSR sensors. Although Sensor 2 produced a peak count closest to the ECG reference, temporal matching was required to determine whether the detected peaks corresponded to the appropriate ECG R-peaks.

For Subject 2, 89 ECG R-peaks were detected. Sensor 1 detected 88 peaks, which was close to the ECG peak count, whereas Sensors 2 and 3 detected only 45 and 55 peaks, respectively. This indicates that Sensor 1 produced a peak count closer to the ECG reference than Sensors 2 and 3. However, temporal matching was required to verify whether these peaks corresponded to the appropriate ECG R-peaks.

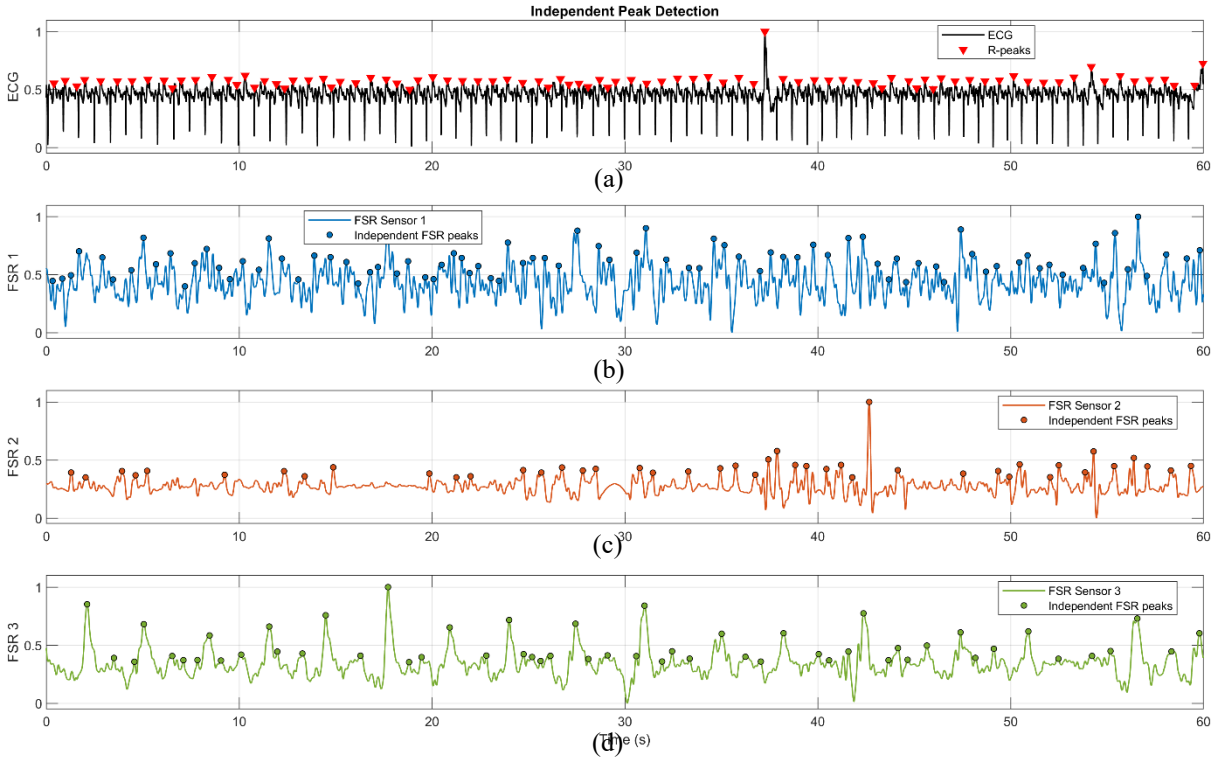


Figure 4.4 Detected ECG and FSR peaks for Subject 2, Trial 4

(a) The raw ECG; (b) Sensor 1; (c) Sensor 2; (d) Sensor 3

As illustrated in Figure 4.4, Sensor 1 produced a greater number of detected peaks than Sensors 2 and 3. Its total of 88 detected peaks was also close to the 89 R-peaks detected from the ECG signal.

For Subject 3, the ECG signal contained 66 detected R-peaks, while Sensors 1, 2, and 3 produced 51, 46, and 54 detected peaks, respectively. All three FSR sensors detected fewer peaks than the ECG reference. This suggests that some mechanical cardiac responses may not have exceeded the selected peak-detection threshold or may have been less distinguishable within the FSR waveforms.

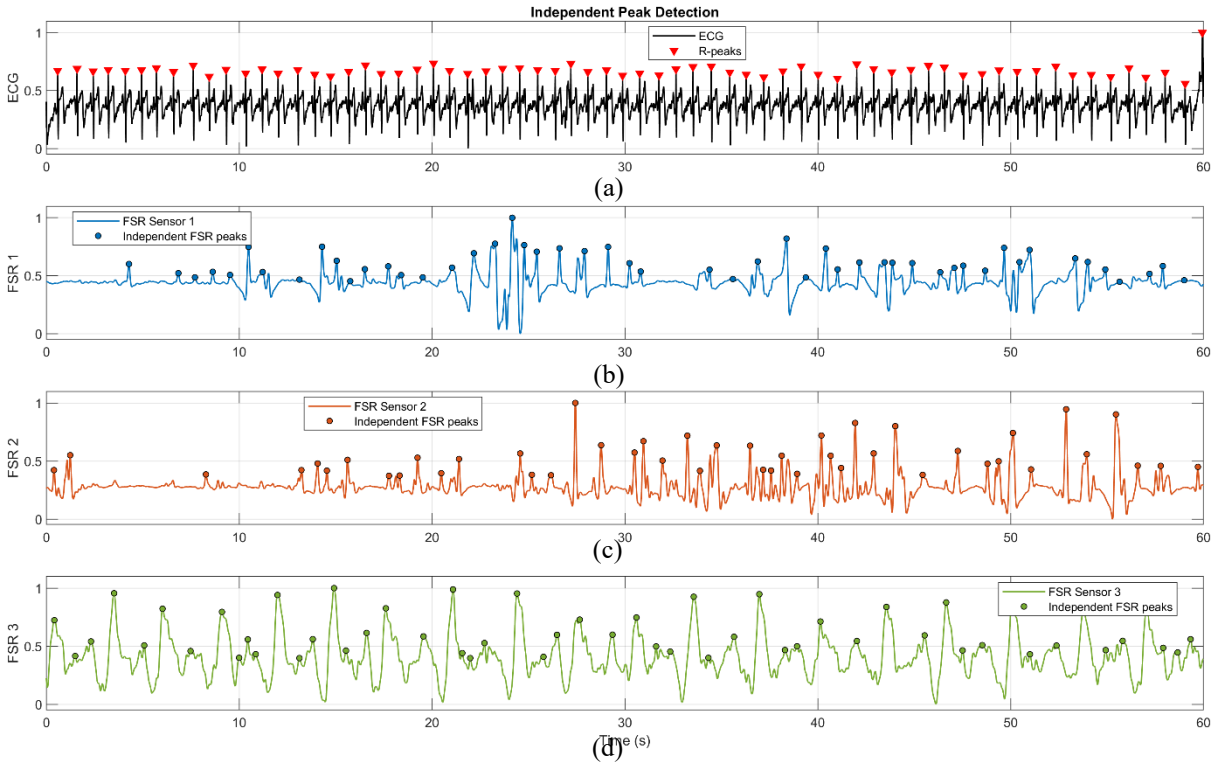


Figure 4.5 Detected ECG and FSR peaks for Subject 3, Trial 3

(a) The raw ECG; (b) Sensor 1; (c) Sensor 2; (d) Sensor 3

Figure 4.5 shows that the number and distribution of the detected FSR peaks differed among the three channels. All three sensors detected fewer peaks than the ECG reference, indicating

that some mechanical responses were not detected using the selected peak-detection parameters.

For Subject 4, 86 ECG R-peaks were detected. Sensors 1 and 3 detected 90 and 83 peaks, respectively, while Sensor 2 detected only 49 peaks. Although the peak counts obtained from Sensors 1 and 3 were relatively close to the ECG peak count, temporal matching was still required to determine whether the detected FSR peaks corresponded to the appropriate ECG R-peaks.

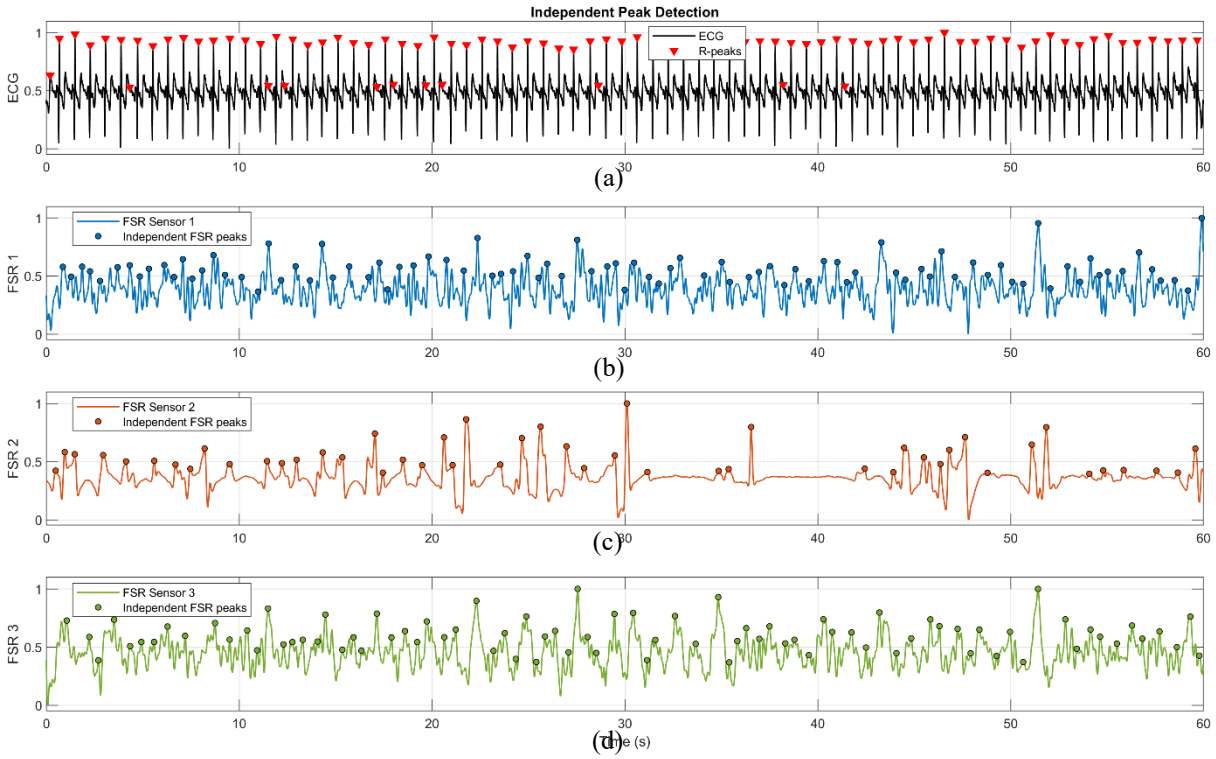


Figure 4.6 Detected ECG and FSR peaks for Subject 4, Trial 3

(a) The raw ECG; (b) Sensor 1; (c) Sensor 2; (d) Sensor 3

As shown in Figure 4.6, Sensors 1 and 3 produced more detected peaks than Sensor 2. Their peak counts were also closer to the ECG reference, although the timing of the peaks still needed to be examined through ECG–FSR matching.

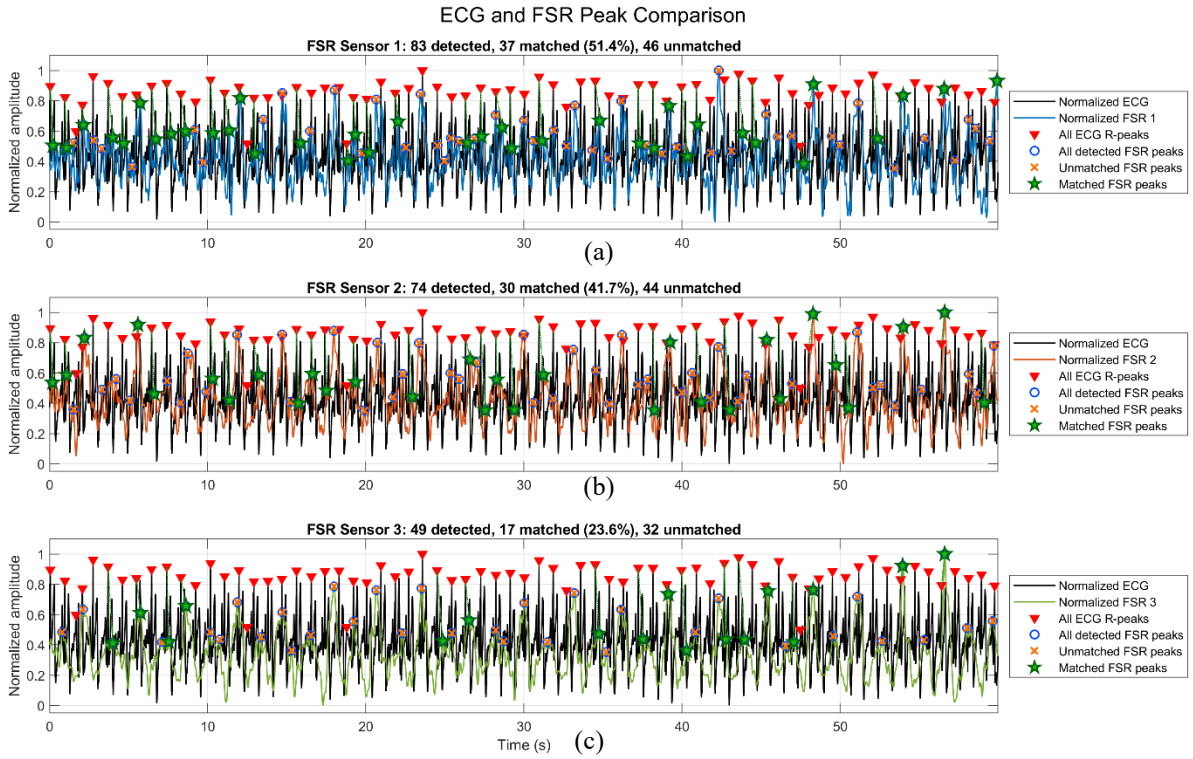


Figure 4.7 ECG and FSR signal overlays with detected peaks for Subject 1, Trial 4

(a) ECG vs Sensor 1 ; (b) ECG vs Sensor 2 ; (c) ECG vs Sensor 3

Figure 4.7 presents the temporal relationship between the ECG R-peaks and the detected peaks from each FSR sensor. The FSR peaks generally occurred after the ECG R-peaks because the signals represented the mechanical cardiac response following the electrical activation measured by the ECG. The ECG–FSR overlay plots for Subjects 2, 3 and 4 are provided in Appendix A.

Overall, cardiac-related mechanical peaks were detected from all three FSR channels across the four subjects. However, the number and distribution of the detected peaks varied between the sensors and subjects. These findings indicate that total peak count alone was insufficient to determine sensor performance. Therefore, the detected peaks were further evaluated using heart rate agreement, temporal peak matching and PTT analysis in the subsequent sections.

4.2.3 Heart Rate Estimation Results

The detected ECG R-peaks and FSR peaks were used to estimate the heart rate for each selected recording. The ECG-derived heart rate served as the reference, while the heart rates obtained from the three FSR sensors were compared with the corresponding ECG value. The performance of each FSR sensor was evaluated using the absolute heart rate error and percentage error, calculated using Absolute HR Error and HR Percentage Error. Table 4.2 presents the ECG and FSR heart rate results for the four selected recordings. The sensor with the smallest absolute heart rate error for each subject is indicated as the best performing sensor in terms of heart rate estimation.

Table 4.2: ECG and FSR heart rate estimation results

Subject	Sensor	ECG HR (bpm)	FSR HR (bpm)	Absolute error (bpm)	Percentage error (%)
1	1	71.31	83.56	12.25	17.18
1	**2**	**71.31**	**76.04**	**4.74**	**6.64**
1	3	71.31	54.41	16.90	23.70
2	**1**	**88.64**	**87.76**	**0.87**	**0.99**
2	2	88.64	68.73	19.91	22.46
2	3	88.64	64.45	24.18	27.28
3	**1**	**65.71**	**64.03**	**1.68**	**2.56**
3	2	65.71	68.18	2.47	3.76
3	3	65.71	61.58	4.14	6.30
4	1	85.74	90.43	4.69	5.47

4	2	85.74	64.48	21.26	24.80
4	**3**	**85.74**	**85.01**	**0.73**	**0.86**

Figure 4.8 presents a representative MATLAB Command Window output for Subject 1, Trial 4. FSR Sensor 2 was selected as the best performing sensor for this recording. A total of 72 ECG R-peaks and 74 FSR peaks were detected. Of these, 30 ECG–FSR peak pairs were successfully matched, resulting in a matched-beat percentage of 41.67%.

```
============================================================
BEST-PERFORMING FSR SENSOR
============================================================
Best-performing sensor: FSR Sensor 2
Detected ECG R-peaks: 72
Detected FSR peaks: 74
Matched ECG-FSR peaks: 30
Matched-beat percentage: 41.67%
ECG heart rate: 71.31 bpm
FSR heart rate: 76.04 bpm
Absolute heart-rate error: 4.74 bpm
Heart-rate percentage error: 6.64%
Mean ECG-to-FSR delay: 185.33 ms
Delay standard deviation: 88.39 ms
============================================================
```

Figure 4.8 Representative MATLAB analysis output for Subject 1, Trial 4

As shown in Figure 4.8, the ECG reference heart rate was 71.31 bpm, while FSR Sensor 2 produced an estimated heart rate of 76.04 bpm. The resulting absolute heart rate error was 4.74 bpm, corresponding to a percentage error of 6.64%. The mean ECG-to-FSR delay was 185.33 ms, with a standard deviation of 88.39 ms. The matched-beat percentage and ECG-to-FSR delay results are discussed further in Section 4.5.

For Subject 1, the reference ECG heart rate was 71.31 bpm. The heart rates estimated by FSR Sensors 1, 2, and 3 were 83.56, 76.04, and 54.41 bpm, respectively. Sensor 2 produced the closest estimate to the ECG reference, with an absolute error of 4.74 bpm and a percentage error of 6.64%. Sensors 1 and 3 produced larger errors of 12.25 and 16.90 bpm, respectively.

For Subject 2, the ECG-derived heart rate was 88.64 bpm. Sensor 1 produced an estimated heart rate of 87.76 bpm, resulting in an absolute error of only 0.87 bpm and a percentage error of 0.99%. In comparison, Sensors 2 and 3 underestimated the reference heart rate, producing percentage errors of 22.46% and 27.28%, respectively. Therefore, Sensor 1 provided the closest heart rate estimate for Subject 2.

For Subject 3, the ECG-derived heart rate was 65.71 bpm. The three FSR sensors produced heart rate estimates of 64.03, 68.18, and 61.58 bpm, respectively. All three estimates were relatively close to the ECG reference compared with the other subjects. Sensor 1 produced the smallest absolute error of 1.68 bpm and a percentage error of 2.56%, followed by Sensor 2 with an error of 2.47 bpm and Sensor 3 with an error of 4.14 bpm.

For Subject 4, the ECG-derived heart rate was 85.74 bpm. Sensor 3 produced the closest estimate at 85.01 bpm, corresponding to an absolute error of 0.73 bpm and a percentage error of 0.86%. Sensor 1 overestimated the reference heart rate by 4.69 bpm, while Sensor 2 underestimated it by 21.26 bpm. Consequently, Sensor 3 provided the best heart rate result for Subject 4.

4.2.4 Pulse Transit Time Analysis

Following peak detection, the ECG R-peaks were matched with the peaks detected from each FSR signal to evaluate their temporal relationship. An FSR peak was considered a valid match when it occurred between 50 and 400 ms after an ECG R-peak. This matching window was applied because the electrical activation recorded by the ECG occurred before the corresponding mechanical cardiac response detected by the FSR sensors. One-to-one matching was used to ensure that each ECG R-peak and FSR peak could only be included in one matched pair. The matched-beat percentage and the time difference between every matched ECG R-peak and FSR peak were calculated using equation (3) and (4). Table 4.3 presents the matched-beat and PTT results for the three FSR sensors across the four selected recordings.

Table 4.3: ECG–FSR matched-beat and PTT results

Subject	Sensor	ECG peaks	Matched beats	Matched beats (%)	Mean PTT (ms)	PTT SD (ms)
1	1	72	37	51.39	185.41	96.86
1	**2**	**72**	**30**	**41.67**	**185.33**	**88.39**
1	3	72	17	23.61	228.24	95.15
2	**1**	**89**	**49**	**55.06**	**246.73**	**98.03**
2	2	89	17	19.10	217.06	111.68
2	3	89	32	35.96	205.94	102.17
3	**1**	**66**	**31**	**46.97**	**199.68**	**99.51**
3	2	66	15	22.73	220.00	83.50
3	3	66	20	30.30	185.00	95.39
4	1	86	41	47.67	183.41	89.18

4	2	86	25	29.07	256.00	89.44
4	**3**	**86**	**39**	**45.35**	**197.18**	**109.64**

The best performing sensor differed among the four subjects. FSR Sensor 2 was selected for Subject 1, FSR Sensor 1 was selected for Subjects 2 and 3, and FSR Sensor 3 was selected for Subject 4. The selection was primarily based on the smallest absolute heart rate error, with PTT variability used when sensors produced equivalent heart rate errors. Therefore, the selected sensor did not necessarily have the highest matched-beat percentage or the lowest PTT standard deviation in every recording. The selected rows are highlighted in bold in Table 4.3.

For Subject 1, Sensor 1 produced the highest matched-beat percentage of 51.39%, with 37 of the 72 ECG R-peaks successfully matched. Sensor 2 produced 30 matched beats, corresponding to 41.67%, while Sensor 3 produced the lowest matched-beat percentage of 23.61%. Although Sensor 1 produced more matched beats, Sensor 2 had the lowest PTT standard deviation of 88.39 ms and the closest FSR heart rate estimate among the three sensors. Therefore, Sensor 2 was selected as the best performing sensor for Subject 1 when the heart rate and timing results were considered together.

For Subject 2, Sensor 1 produced the strongest peak-matching result, with 49 of the 89 ECG R-peaks successfully matched. This corresponded to the highest matched-beat percentage of 55.06% among all the selected recordings. Sensor 1 also produced a mean PTT of 246.73 ms and a PTT standard deviation of 98.03 ms. Sensors 2 and 3 achieved lower matched-beat percentages of 19.10% and 35.96%, respectively. These findings were consistent with the heart rate results, in which Sensor 1 produced the closest estimate to the ECG reference.

For Subject 3, Sensor 1 produced the highest matched-beat percentage of 46.97%, with 31 of the 66 ECG R-peaks successfully matched. Sensors 2 and 3 produced matched-beat percentages of 22.73% and 30.30%, respectively. Sensor 2 produced the lowest PTT standard deviation of 83.50 ms, but only 15 ECG–FSR peak pairs were matched. Therefore, its lower variability was calculated from a smaller number of matched beats and should not be considered independently. Sensor 1 provided the strongest overall balance between matched-beat count and heart rate agreement for this subject.

For Subject 4, Sensors 1 and 3 produced similar matching results. Sensor 1 matched 41 of the 86 ECG R-peaks, corresponding to 47.67%, while Sensor 3 matched 39 peaks, corresponding to 45.35%. Although Sensor 1 produced two more matched beats and a lower PTT standard deviation, Sensor 3 produced an FSR heart rate of 85.01 bpm, which was closest to the ECG reference of 85.74 bpm. Sensor 3 was therefore selected as the best performing sensor when both heart rate accuracy and peak-matching performance were considered.

The mean PTT values across the four subjects and three sensors ranged from 183.41 to 256.00 ms. All mean values remained within the predefined matching interval of 50–400 ms. However, the PTT standard deviations ranged from 83.50 to 111.68 ms, indicating that the delay between the ECG R-peaks and FSR peaks varied during the recordings. This variability may have been caused by differences in waveform shape, sensor position, pressure distribution and the transmission of mechanical cardiac activity through the mattress.

As illustrated in Figure 4.7, only FSR peaks occurring within the specified 50–400 ms interval after an ECG R-peak were accepted as matched cardiac events. The matched peaks were subsequently used to calculate the mean PTT and PTT standard deviation.

Overall, the results demonstrate that valid temporal relationships could be identified between the electrical cardiac activity recorded by the ECG and the mechanical responses detected by the FSR sensors. Nevertheless, the matched-beat percentage and PTT variability differed among sensors and subjects. These differences are considered together with heart rate accuracy in Section 4.6 to compare the performance of the three FSR sensors.

4.2.5 Comparison Of FSR Sensor Performance Across Subjects

The performance of the three FSR sensors was compared across the four selected recordings using the mean absolute heart rate error, matched-beat percentage and PTT variability. Unlike the previous sections, which discussed each subject individually, this section evaluates whether any sensor demonstrated consistently stronger performance across the subjects. Table 4.4 presents the average performance of each sensor across the four subjects.

Table 4.4: Average performance of the three FSR sensors across the four subjects

Sensor	Mean absolute HR error (bpm)	Mean matched beats (%)	Mean PTT SD (ms)	Number of times selected as best
Sensor 1	4.87	50.27	93.39	2
Sensor 2	12.09	28.14	93.25	1
Sensor 3	11.49	33.80	100.59	1

FSR Sensor 1 demonstrated the strongest overall performance across the four subjects. It produced the lowest mean absolute heart rate error of 4.87 bpm and the highest mean matched-beat percentage of 50.27%. It was also selected as the best performing sensor for Subjects 2 and 3. In comparison, Sensors 2 and 3 were each selected as the best performing sensor for one subject.

Sensor 2 produced the lowest average PTT standard deviation of 93.25 ms, although the difference from Sensor 1 which produced 93.39 ms was small. However, Sensor 2 had the lowest mean matched-beat percentage and the highest mean heart rate error. Sensor 3 produced a higher mean matched-beat percentage than Sensor 2 but had the highest average PTT variability.

Despite the stronger overall performance of Sensor 1, no single sensor performed best for all four subjects. Sensor 2 was selected for Subject 1, Sensor 1 for Subjects 2 and 3, and Sensor 3 for Subject 4. This variation may have resulted from differences in thoracic alignment, body dimensions, weight and pressure distribution across the sensor array. Although the sensor-placement distance was adjusted according to each subject's height, the amount of mechanical cardiac activity transferred to each sensor could still differ.

These findings support the use of a three-sensor array rather than a single fixed FSR sensor. The array increased the likelihood that at least one sensor would receive a sufficiently clear mechanical cardiac signal for heart rate estimation. Nevertheless, the results were obtained from only four subjects in the supine position. Therefore, Sensor 1 can be identified as the strongest-performing sensor within the present experiment, but it should not be

considered universally optimal without evaluation involving a larger and more diverse group of subjects.

4.2.6 Factors Affecting FSR Sensor Performance And Possible Improvements

Although cardiac-related peaks were detected from all three FSR sensors, the matched-beat percentages ranged from 19.10% to 55.06%. This indicates that not every ECG R-peak produced a clearly detectable and correctly matched FSR peak. The variation may have resulted from differences in body structure, weight, thoracic alignment and pressure distribution across the sensor array. Although the placement distance was adjusted according to each subject's height, the thoracic region may not have been centred over the same sensor for every subject. Insufficient sensor loading could produce weak cardiac-related changes, while excessive loading could reduce the sensor's responsiveness to small pressure variations. The mattress may also have attenuated or distributed the cardiac-induced vibrations before they reached the sensors, while differences in clothing could have affected the mechanical coupling between the subject and mattress. The subjects wore clothing made from different materials and of different thicknesses during data collection. This lack of clothing standardisation may have affected the transmission of mechanical cardiac activity from the body to the mattress. Thick, loose or highly elastic materials may absorb or distribute small body vibrations differently from thin and uniform lightweight materials. In a hospital environment, patients normally wear relatively thin and more standardised hospital garments, which may provide more consistent mechanical coupling between the body and mattress. However, since the effect of clothing material was not tested independently in this study, it can only be considered a possible contributing factor.

Respiration and small involuntary movements may have introduced additional variations into the FSR signals. Breathing produces larger and slower thoracic movements than cardiac activity. Some respiratory influence may have remained near the lower cutoff frequency of the 0.5–5 Hz bandpass filter. Muscle tension, swallowing and slight repositioning could also have produced fluctuations resembling cardiac peaks. Electronic noise, power-supply variations, connection resistance, the limited resolution of the Arduino Uno analogue-to-digital converter, and the nonlinear response and hysteresis of the FSR-408 sensors may have further reduced signal consistency. In addition, the fixed minimum peak height of 0.50 may have excluded some low-amplitude cardiac peaks while accepting unrelated fluctuations. Missed peaks could underestimate the FSR heart rate, whereas additional false peaks could cause it to be overestimated.

The system could be improved by using adaptive peak-detection parameters based on the noise level and waveform characteristics of each FSR channel. Methods such as moving-window thresholding, peak-prominence analysis or template-based detection could help distinguish cardiac peaks from respiratory and movement-related fluctuations. The three FSR signals could also be combined so that events detected at approximately the same time by multiple sensors are accepted with greater confidence. This approach could reduce false detections and compensate when one sensor receives a weak mechanical signal.

Further mechanical and electronic improvements could include applying a more uniform preload across the sensors, adding a thin pressure-distribution layer and investigating different sensor spacing and mattress characteristics. Operational-amplifier buffers, decoupling capacitors, stable power regulation, proper grounding, shielded wires and a higher-resolution external analogue-to-digital converter could help reduce electronic noise. An accelerometer

could also be incorporated to identify movement-contaminated signal segments. Future testing should involve more subjects, controlled clothing conditions, different mattress characteristics and additional sleeping positions. These improvements may increase the matched-beat percentage, reduce heart rate error and produce more consistent PTT measurements.

CHAPTER 5

CONCLUSION

5.1 CONCLUSION

This project successfully developed a bed-based unobtrusive cardiac monitoring system using a three-sensor FSR array. The first objective was to design and develop a non-invasive sensor pad capable of detecting cardiac-related mechanical activity for bedridden patients. This objective was fully achieved through the development of a sensor pad comprising three FSR-408 sensors connected to an Arduino Uno using voltage-divider circuits. The pad was positioned on top of the mattress beneath the subject at the thoracic region, allowing cardiac-induced mechanical activity to be recorded without directly attaching the FSR sensors to the subject. The prototype successfully acquired cardiac-related FSR signals from all four subjects while they maintained the supine position.

The second objective was to analyse signal-conditioning and signal-processing methods for extracting heart-rate data from the signals captured by the sensor pad. This objective was fully achieved through the development of a MATLAB-based processing procedure involving signal synchronisation, trimming, bandpass filtering, Min–Max normalisation and peak detection. The detected FSR peaks were used to estimate heart rate and were matched with the ECG R-peaks within a 50–400 ms interval. The method successfully produced FSR-derived heart-rate estimates for all four subjects. However, the matched-beat percentages ranged from 19.10% to 55.06%, showing that further improvement is required to achieve more consistent beat-by-beat detection.

The third objective was to evaluate the accuracy of the developed system by comparing the FSR-derived heart rate with a standard ECG reference device. This objective was fully achieved by comparing the FSR measurements with ECG signals recorded simultaneously using the BITalino device. The best-performing sensors for Subjects 1, 2, 3 and 4 produced absolute heart-rate errors of 4.74, 0.87, 1.68 and 0.73 bpm, respectively. Sensor 1 demonstrated the strongest overall performance, with a mean absolute heart-rate error of 4.87 bpm and a mean matched-beat percentage of 50.27%. Nevertheless, the best-performing sensor differed among the subjects, supporting the use of a three-sensor array rather than a single fixed sensor. Overall, all three research objectives were achieved, demonstrating the feasibility of using an FSR sensor array beneath a mattress for non-invasive heart-rate monitoring. However, further improvement and testing are required before the system can be considered suitable for reliable continuous or clinical monitoring.

5.2 FUTURE WORKS

Future work should prioritise the detection and reduction of motion artefacts caused by respiration, repositioning and small involuntary body movements. An accelerometer could be incorporated to identify movement-contaminated periods, while adaptive filtering and peak-detection thresholds could improve the separation of cardiac activity from noise. Clothing should also be standardised during future experiments, preferably using thin hospital type garments, to reduce variations caused by different fabric materials and thicknesses. Further improvements may include combining information from all three FSR sensors, improving the signal-conditioning circuit, using a higher-resolution analogue-to-digital converter and providing more uniform loading across the sensor array. The system should subsequently be evaluated with more subjects, different mattresses and additional sleeping positions to determine its reliability under a wider range of conditions.

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APPENDIX A

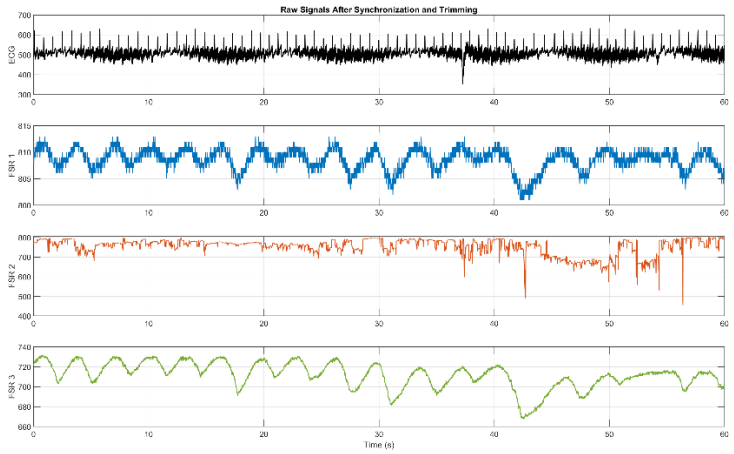


Figure A.1 Trimmed Raw ECG and FSR Signals for Subject 2

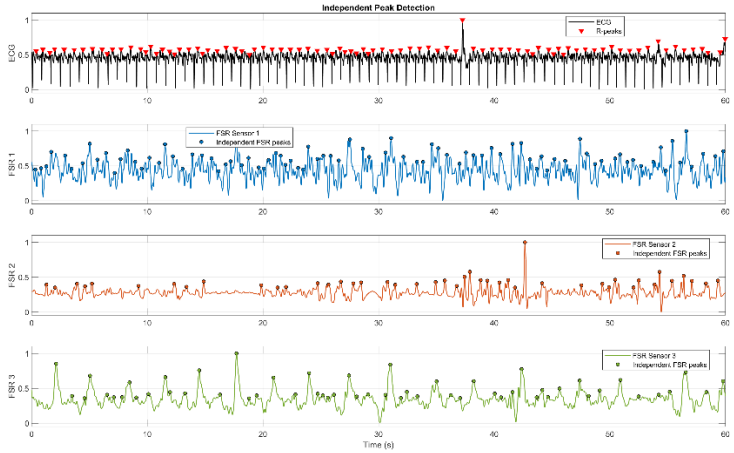


Figure A.2 Filtered and Normalised ECG and FSR Signals for Subject 2

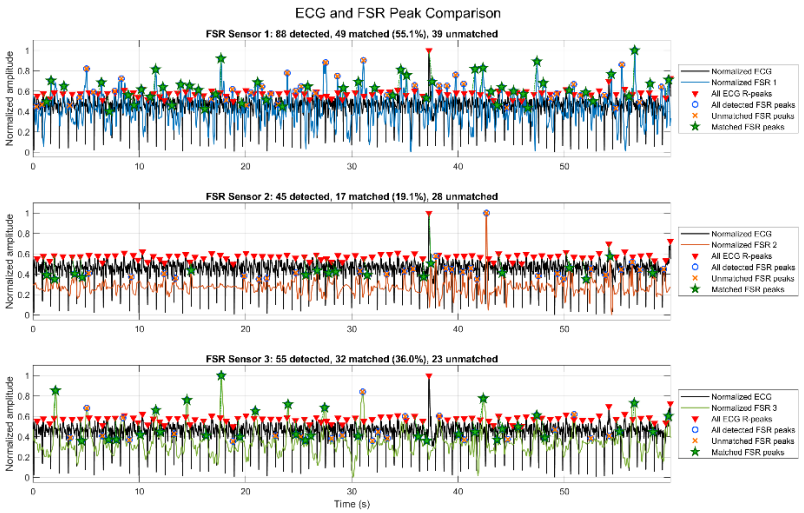


Figure A.3 ECG and FSR Signal Overlays for Subject 2

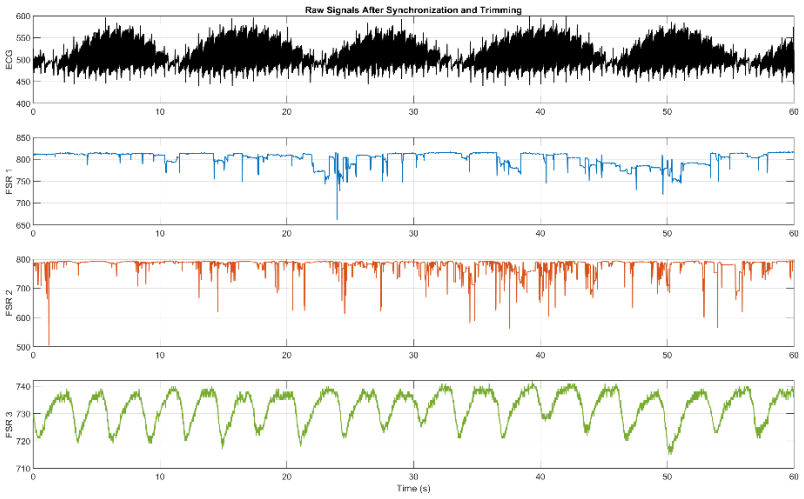


Figure A.4 Trimmed Raw ECG and FSR Signals for Subject 3

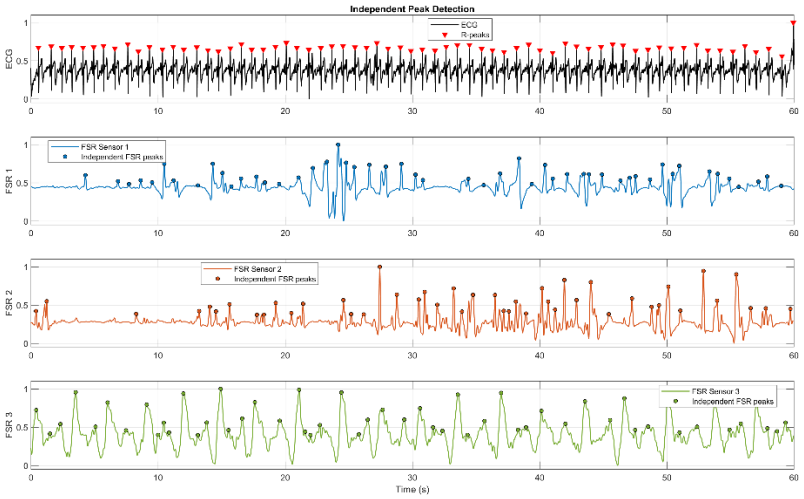


Figure A.5 Filtered and Normalised ECG and FSR Signals for Subject 3

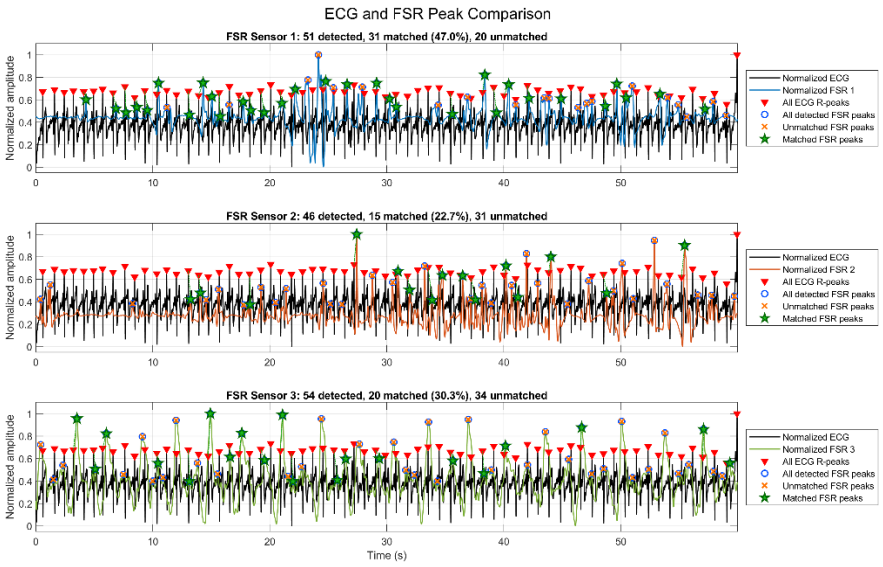


Figure A.6 ECG and FSR Signal Overlays for Subject 3

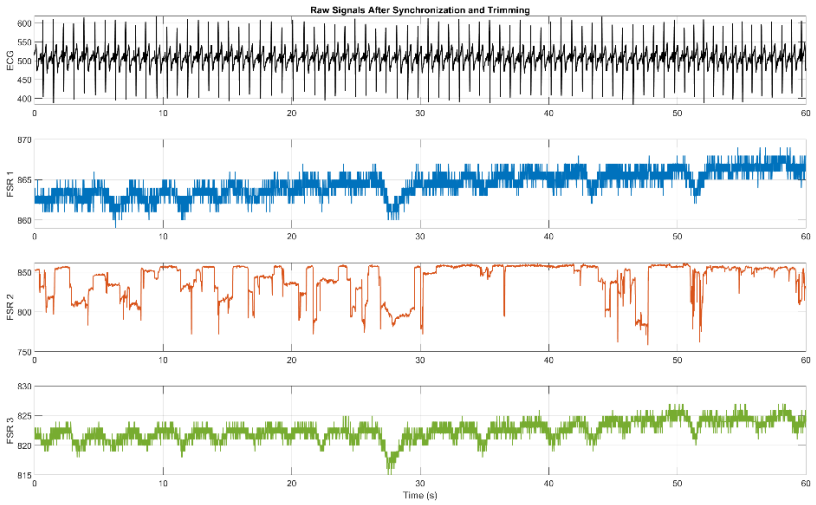


Figure A.7 Trimmed Raw ECG and FSR Signals for Subject 4

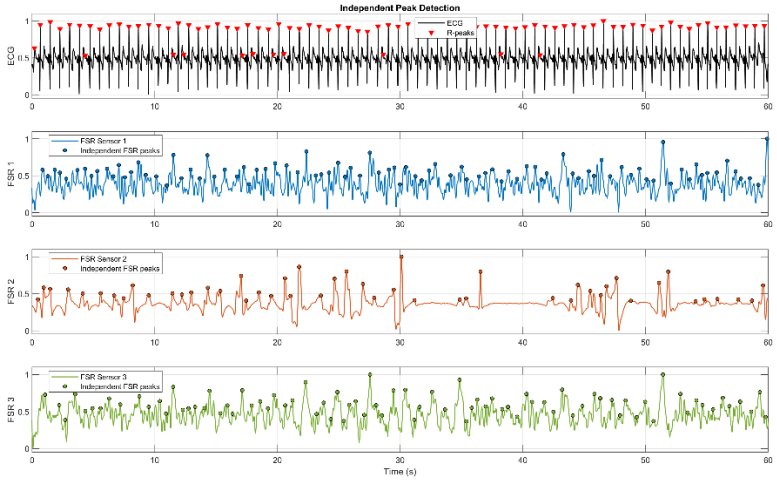


Figure A.8 Filtered and Normalised ECG and FSR Signals for Subject 4

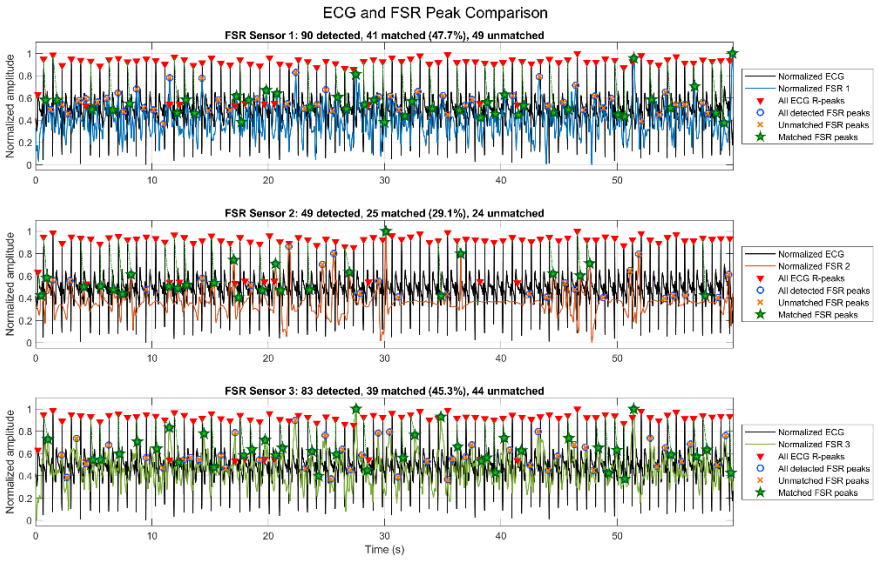


Figure A.9 ECG and FSR Signal Overlays for Subject 4

APPENDIX B

Listing B.1 MATLAB Code to Acquire Data from FSR

```
%% Record Three-Channel FSR Data
clear; clc;

portName = "COM4";
baudRate = 115200;

s = serialport(portName, baudRate);
configureTerminator(s, "LF");

duration = 80;
fs = 100;
numSamples = duration * fs;
numSensors = 3;

disp("Recording started... Please stay still.");

data = zeros(numSamples, numSensors);

tic;
for i = 1:numSamples
    lineStr = readline(s);
    parsedValues = str2double(split(lineStr, sprintf('\t')));

    if length(parsedValues) >= numSensors
        data(i, :) = parsedValues(1:numSensors)';
    end

    if mod(i, fs * 5) == 0
        elapsed_sec = i / fs;
        fprintf('Progress: %.0f / %d seconds recorded...\n', ...
            elapsed_sec, duration);
    end
end
toc;

clear s;

fsr_raw = data;
save('reenfsr_data.mat', 'fsr_raw');
disp("Recording finished and saved as 'iw_fsr_data.mat'");
```

Listing B.2 MATLAB Code for ECG and FSR Signal Processing and Comparative Analysis

```
clear;
clc;
```

```
close all;
%% 1. USER SETTINGS
fs_fsr = 100;                    % FSR sampling frequency (Hz)
fs_ecg = 100;                    % ECG sampling frequency (Hz)
target_duration = 80;              % Final 80 seconds selected
trim_start = 10;                 % Remove first 10 seconds
trim_end = 10;                   % Remove final 10 seconds
fsr_filename = 'reen4.mat';
ecg_filename = 'reen4.txt';
[fsr_folder,fsr_base_name,~] = fileparts(fsr_filename);
if isempty(fsr_folder)
    figure_output_folder = pwd;
else
    figure_output_folder = fsr_folder;
end
ecg_column = [];
fsr_low_cutoff = 0.5;
fsr_high_cutoff = 5.0;
ecg_low_cutoff = 0.5;
ecg_high_cutoff = 40.0;
filter_order = 4;
minimum_beat_distance = 0.40;       % Minimum distance = 0.40 s
ecg_min_peak_height = 0.50;         % Normalized ECG threshold
fsr_min_peak_height = 0.35;         % Normalized FSR threshold
fsr_min_peak_prominence = 0.05;     % FSR prominence requirement
min_delay = 0.05;                  % 50 ms
max_delay = 0.40;                  % 400 ms
ecg_direction = 0;
fsr_direction = [0 0 0];
%% 2. LOAD FSR DATA
if ~isfile(fsr_filename)
    error('The FSR file "%s" was not found.',fsr_filename);
end
fsr_file = load(fsr_filename);
if ~isfield(fsr_file,'fsr_raw')
    error(['The FSR file does not contain a variable named fsr_raw. ', ...
          'Check the FSR recording code.']);
end
fsr_raw = double(fsr_file.fsr_raw);
if size(fsr_raw,1) == 3 && size(fsr_raw,2) > 3
    fsr_raw = fsr_raw.';
    warning('FSR data were transposed from 3-by-N to N-by-3.');
end
if size(fsr_raw,2) ~= 3
    error(['fsr_raw must contain exactly three columns: ', ...
          'Sensor 1, Sensor 2 and Sensor 3.']);
end
fprintf('============================================================\n');
fprintf('FSR DATA INFORMATION\n');
fprintf('============================================================\n');
```

```
fprintf('FSR file: %s\n',fsr_filename);
fprintf('FSR samples: %d\n',size(fsr_raw,1));
fprintf('FSR sensors: %d\n',size(fsr_raw,2));
fprintf('FSR duration based on 100 Hz: %.2f seconds\n\n', ...
   size(fsr_raw,1)/fs_fsr);
%% 3. CHECK AND REPAIR INVALID FSR SERIAL READINGS
invalid_fsr_rows = ...
   ~all(isfinite(fsr_raw),2) | ...
   all(fsr_raw == 0,2);
number_invalid_fsr = sum(invalid_fsr_rows);
if number_invalid_fsr > 0
   warning('%d invalid FSR rows were detected.', ...
      number_invalid_fsr);
   fsr_raw(invalid_fsr_rows,:) = NaN;
   fsr_raw = fillmissing(fsr_raw,'linear',1);
   fsr_raw = fillmissing(fsr_raw,'nearest',1);
   if any(~isfinite(fsr_raw(:)))
      error('Some invalid FSR readings could not be repaired.');
   end
   fprintf('%d invalid FSR rows were repaired.\n\n', ...
      number_invalid_fsr);
else
   fprintf('No invalid FSR serial rows were found.\n\n');
end
expected_fsr_samples = round(target_duration*fs_fsr);
if size(fsr_raw,1) ~= expected_fsr_samples
   warning(['Expected %d FSR samples for %.1f seconds, ', ...
         'but %d samples were found.'], ...
         expected_fsr_samples, ...
         target_duration, ...
         size(fsr_raw,1));
end
%% 4. DISPLAY BASIC FSR INFORMATION
fsr_minimum = min(fsr_raw,[],1);
fsr_maximum = max(fsr_raw,[],1);
fsr_raw_range = fsr_maximum-fsr_minimum;
fsr_mean = mean(fsr_raw,1);
fsr_standard_deviation = std(fsr_raw,0,1);
fsr_information = table( ...
   (1:3)', ...
   fsr_minimum', ...
   fsr_maximum', ...
   fsr_raw_range', ...
   fsr_mean', ...
   fsr_standard_deviation', ...
   'VariableNames',{ ...
   'FSR_Sensor', ...
   'Minimum_Raw_Value', ...
   'Maximum_Raw_Value', ...
   'Raw_Range', ...
```

```
    'Raw_Mean', ...
    'Raw_Standard_Deviation'});
disp(fsr_information);
%% 5. LOAD BITALINO ECG DATA
if ~isfile(ecg_filename)
    error('The ECG file "%s" was not found.',ecg_filename);
end
try
    import_options = detectImportOptions( ...
        ecg_filename, ...
        'FileType','text');
    import_options.CommentStyle = '#';
    ecg_table = readtable( ...
        ecg_filename, ...
        import_options);
    ecg_numeric = table2array(ecg_table);
catch
    error(['The ECG file could not be imported. Confirm that it is ', ...
          'a BITalino/OpenSignals numerical TXT file.']);
end
if ~isnumeric(ecg_numeric) || isempty(ecg_numeric)
    error('No usable numerical ECG data were found.');
end
ecg_numeric = double(ecg_numeric);
if isempty(ecg_column)
    ecg_column = size(ecg_numeric,2);
end
if ecg_column < 1 || ecg_column > size(ecg_numeric,2)
    error(['ECG column %d does not exist. The file contains ', ...
          '%d numerical columns.'], ...
          ecg_column, ...
          size(ecg_numeric,2));
end
ecg_raw = ecg_numeric(:,ecg_column);
%% 6. CHECK AND REPAIR INVALID ECG READINGS
invalid_ecg_samples = ~isfinite(ecg_raw);
number_invalid_ecg = sum(invalid_ecg_samples);
if number_invalid_ecg > 0
    warning('%d invalid ECG samples were detected.', ...
        number_invalid_ecg);
    ecg_raw(invalid_ecg_samples) = NaN;
    ecg_raw = fillmissing(ecg_raw,'linear');
    ecg_raw = fillmissing(ecg_raw,'nearest');
    if any(~isfinite(ecg_raw))
        error('Some invalid ECG readings could not be repaired.');
    end
    fprintf('%d invalid ECG samples were repaired.\n', ...
        number_invalid_ecg);
end
fprintf('\n=========================================================\n');
```

```
fprintf('ECG DATA INFORMATION\n');
fprintf('============================================================\n');
fprintf('ECG file: %s\n',ecg_filename);
fprintf('Selected ECG column: %d\n',ecg_column);
fprintf('ECG samples: %d\n',length(ecg_raw));
fprintf('ECG duration: %.2f seconds\n\n', ...
   length(ecg_raw)/fs_ecg);
%% 7. SELECT THE FINAL 80 SECONDS
required_fsr_samples = round(target_duration*fs_fsr);
required_ecg_samples = round(target_duration*fs_ecg);
if size(fsr_raw,1) >= required_fsr_samples
   fsr_selected = fsr_raw( ...
      end-required_fsr_samples+1:end,:);
else
   warning(['FSR recording is shorter than %.1f seconds. ', ...
         'The complete available FSR recording will be used.'], ...
         target_duration);
   fsr_selected = fsr_raw;
end
if length(ecg_raw) >= required_ecg_samples
   ecg_selected = ecg_raw( ...
      end-required_ecg_samples+1:end);
else
   warning(['ECG recording is shorter than %.1f seconds. ', ...
         'The complete available ECG recording will be used.'], ...
         target_duration);
   ecg_selected = ecg_raw;
end
fprintf('Selected FSR duration: %.2f seconds\n', ...
   size(fsr_selected,1)/fs_fsr);
fprintf('Selected ECG duration: %.2f seconds\n\n', ...
   length(ecg_selected)/fs_ecg);
%% 8. SYNCHRONIZE FSR AND ECG FROM THEIR ENDING POINTS
if fs_fsr ~= fs_ecg
   error(['This code requires ECG and FSR to have the same ', ...
        'sampling frequency.']);
end
common_samples = min( ...
   size(fsr_selected,1), ...
   length(ecg_selected));
fsr_synchronized = fsr_selected( ...
   end-common_samples+1:end,:);
ecg_synchronized = ecg_selected( ...
   end-common_samples+1:end);
synchronized_duration = common_samples/fs_fsr;
fprintf('============================================================\n');
fprintf('SYNCHRONIZATION INFORMATION\n');
fprintf('============================================================\n');
fprintf('Common samples: %d\n',common_samples);
fprintf('Duration before trimming: %.2f seconds\n\n', ...
```

```
    synchronized_duration);
%% 9. REMOVE FIRST AND FINAL 10 SECONDS
trim_start_samples = round(trim_start*fs_fsr);
trim_end_samples = round(trim_end*fs_fsr);
first_sample = trim_start_samples+1;
last_sample = common_samples-trim_end_samples;
if last_sample <= first_sample
    error(['The synchronized data are too short for the requested ', ...
           'start and end trimming.']);
end
fsr_trimmed = fsr_synchronized(first_sample:last_sample,:);
ecg_trimmed = ecg_synchronized(first_sample:last_sample);
number_of_samples = length(ecg_trimmed);
analysis_duration = number_of_samples/fs_ecg;
time = (0:number_of_samples-1)'/fs_ecg;
fprintf('Removed from beginning: %.2f seconds\n',trim_start);
fprintf('Removed from end: %.2f seconds\n',trim_end);
fprintf('Remaining samples: %d\n',number_of_samples);
fprintf('Final analysis duration: %.2f seconds\n\n', ...
    analysis_duration);
%% 10. REMOVE MEAN AND LINEAR TREND
ecg_detrended = detrend(ecg_trimmed);
fsr_detrended = zeros(size(fsr_trimmed));
for sensor = 1:3
    fsr_detrended(:,sensor) = ...
        detrend(fsr_trimmed(:,sensor));
end
%% 11. BANDPASS FILTERING
if fsr_high_cutoff >= fs_fsr/2
    error('FSR upper cutoff must be below the Nyquist frequency.');
end
if ecg_high_cutoff >= fs_ecg/2
    error('ECG upper cutoff must be below the Nyquist frequency.');
end
[fsr_b,fsr_a] = butter( ...
    filter_order/2, ...
    [fsr_low_cutoff fsr_high_cutoff]/(fs_fsr/2), ...
    'bandpass');
[ecg_b,ecg_a] = butter( ...
    filter_order/2, ...
    [ecg_low_cutoff ecg_high_cutoff]/(fs_ecg/2), ...
    'bandpass');
ecg_filtered = filtfilt( ...
    ecg_b, ...
    ecg_a, ...
    ecg_detrended);
fsr_filtered = zeros(size(fsr_detrended));
for sensor = 1:3
    fsr_filtered(:,sensor) = filtfilt( ...
        fsr_b, ...
```

```
        fsr_a, ...
        fsr_detrended(:,sensor));
end
%% 12. CORRECT SIGNAL DIRECTION AND NORMALIZE
%% Correct ECG direction
if ecg_direction == -1
    ecg_filtered = -ecg_filtered;
elseif ecg_direction == 0
    if abs(min(ecg_filtered)) > abs(max(ecg_filtered))
        ecg_filtered = -ecg_filtered;
    end
end
%% Correct FSR directions
for sensor = 1:3
    if fsr_direction(sensor) == -1
        fsr_filtered(:,sensor) = ...
            -fsr_filtered(:,sensor);
    elseif fsr_direction(sensor) == 0
        if abs(min(fsr_filtered(:,sensor))) > ...
                abs(max(fsr_filtered(:,sensor)))
            fsr_filtered(:,sensor) = ...
                -fsr_filtered(:,sensor);
        end
    end
end
%% Normalize every signal independently between 0 and 1
ecg_normalized = normalizeMinMax(ecg_filtered);
fsr_normalized = zeros(size(fsr_filtered));
for sensor = 1:3
    fsr_normalized(:,sensor) = ...
        normalizeMinMax(fsr_filtered(:,sensor));
end
%% 13. DETECT ECG R-PEAKS INDEPENDENTLY
minimum_ecg_peak_samples = ...
    round(minimum_beat_distance*fs_ecg);
[ecg_peak_values,ecg_peak_locations] = findpeaks( ...
    ecg_normalized, ...
    'MinPeakHeight',ecg_min_peak_height, ...
    'MinPeakDistance',minimum_ecg_peak_samples);
ecg_peak_times = time(ecg_peak_locations);
number_of_ecg_beats = length(ecg_peak_locations);
fprintf('============================================================\n');
fprintf('INDEPENDENT PEAK DETECTION\n');
fprintf('============================================================\n');
fprintf('ECG R-peaks detected: %d\n',number_of_ecg_beats);
%% 14. CALCULATE ECG HEART RATE
ecg_heart_rate = NaN;
if length(ecg_peak_times) >= 2
    rr_intervals = diff(ecg_peak_times);
    valid_rr = ...
```

```
        rr_intervals >= 0.40 & ...
        rr_intervals <= 1.50;
    rr_intervals = rr_intervals(valid_rr);
    if ~isempty(rr_intervals)
        ecg_heart_rate = 60/mean(rr_intervals);
    end
end
fprintf('Estimated ECG heart rate: %.2f bpm\n\n', ...
    ecg_heart_rate);
%% 15. DETECT FSR PEAKS INDEPENDENTLY
minimum_fsr_peak_samples = ...
    round(minimum_beat_distance*fs_fsr);
fsr_peak_values = cell(3,1);
fsr_peak_locations = cell(3,1);
fsr_peak_times = cell(3,1);
for sensor = 1:3
    [fsr_peak_values{sensor}, ...
     fsr_peak_locations{sensor}] = findpeaks( ...
        fsr_normalized(:,sensor), ...
        'MinPeakHeight',fsr_min_peak_height, ...
        'MinPeakProminence',fsr_min_peak_prominence, ...
        'MinPeakDistance',minimum_fsr_peak_samples);
    fsr_peak_times{sensor} = ...
        time(fsr_peak_locations{sensor});
    fprintf('FSR Sensor %d independently detected peaks: %d\n', ...
        sensor,length(fsr_peak_locations{sensor}));
end
fprintf('\n');
%% 16. MATCH THE INDEPENDENT ECG AND FSR PEAKS
matched_ecg_times = cell(3,1);
matched_fsr_times = cell(3,1);
matched_ecg_indices = cell(3,1);
matched_fsr_indices = cell(3,1);
delay_seconds = cell(3,1);
for sensor = 1:3
    current_fsr_times = ...
        fsr_peak_times{sensor};
    current_fsr_locations = ...
        fsr_peak_locations{sensor};
    current_fsr_values = ...
        fsr_peak_values{sensor};
    used_fsr_peak = ...
        false(length(current_fsr_times),1);
    sensor_ecg_times = [];
    sensor_fsr_times = [];
    sensor_ecg_indices = [];
    sensor_fsr_indices = [];
    sensor_delays = [];
    for beat = 1:number_of_ecg_beats
        candidate_delays = ...
```

```
            current_fsr_times-ecg_peak_times(beat);
        candidate_indices = find( ...
            candidate_delays >= min_delay & ...
            candidate_delays <= max_delay & ...
            ~used_fsr_peak);
        if ~isempty(candidate_indices)
            candidate_heights = ...
                current_fsr_values(candidate_indices);
            [~,strongest_position] = ...
                max(candidate_heights);
            selected_candidate = ...
                candidate_indices(strongest_position);
            used_fsr_peak(selected_candidate) = true;
            sensor_ecg_times(end+1,1) = ...
                ecg_peak_times(beat);
            sensor_fsr_times(end+1,1) = ...
                current_fsr_times(selected_candidate);
            sensor_ecg_indices(end+1,1) = ...
                ecg_peak_locations(beat);
            sensor_fsr_indices(end+1,1) = ...
                current_fsr_locations(selected_candidate);
            sensor_delays(end+1,1) = ...
                candidate_delays(selected_candidate);
        end
    end
    matched_ecg_times{sensor} = ...
        sensor_ecg_times;
    matched_fsr_times{sensor} = ...
        sensor_fsr_times;
    matched_ecg_indices{sensor} = ...
        sensor_ecg_indices;
    matched_fsr_indices{sensor} = ...
        sensor_fsr_indices;
    delay_seconds{sensor} = ...
        sensor_delays;
end
%% 17. CALCULATE HEART-RATE AND TIMING RESULTS
total_fsr_peaks = zeros(3,1);
matched_beats = zeros(3,1);
matched_beat_percentage = nan(3,1);
unmatched_fsr_peaks = zeros(3,1);
unmatched_ecg_peaks = zeros(3,1);
fsr_heart_rate = nan(3,1);
heart_rate_error = nan(3,1);
heart_rate_percentage_error = nan(3,1);
mean_delay_ms = nan(3,1);
std_delay_ms = nan(3,1);
for sensor = 1:3
    total_fsr_peaks(sensor) = ...
        length(fsr_peak_locations{sensor});
```

```
    matched_beats(sensor) = ...
        length(delay_seconds{sensor});
    unmatched_fsr_peaks(sensor) = ...
        max(total_fsr_peaks(sensor)-matched_beats(sensor),0);
    unmatched_ecg_peaks(sensor) = ...
        max(number_of_ecg_beats-matched_beats(sensor),0);
    if number_of_ecg_beats > 0
        matched_beat_percentage(sensor) = ...
            100*matched_beats(sensor)/number_of_ecg_beats;
    end
    if ~isempty(delay_seconds{sensor})
        mean_delay_ms(sensor) = ...
            mean(delay_seconds{sensor})*1000;
        std_delay_ms(sensor) = ...
            std(delay_seconds{sensor})*1000;
    end
    if length(fsr_peak_times{sensor}) >= 2
        fsr_intervals = ...
            diff(fsr_peak_times{sensor});
        valid_fsr_intervals = ...
            fsr_intervals >= 0.40 & ...
            fsr_intervals <= 1.50;
        fsr_intervals = ...
            fsr_intervals(valid_fsr_intervals);
        if ~isempty(fsr_intervals)
            fsr_heart_rate(sensor) = ...
                60/mean(fsr_intervals);
        end
    end
    if isfinite(fsr_heart_rate(sensor)) && ...
            isfinite(ecg_heart_rate)
        heart_rate_error(sensor) = ...
            abs(fsr_heart_rate(sensor)-ecg_heart_rate);
        if ecg_heart_rate > 0
            heart_rate_percentage_error(sensor) = ...
                100*heart_rate_error(sensor)/ecg_heart_rate;
        end
    end
end
%% 18. SELECT THE BEST FSR SENSOR
valid_sensors = isfinite(heart_rate_error);
if ~any(valid_sensors)
    best_sensor = NaN;
else
    minimum_hr_error = ...
        min(heart_rate_error(valid_sensors));
    sensor_candidates = find( ...
        valid_sensors & ...
        abs(heart_rate_error-minimum_hr_error) < 1e-10);
    if length(sensor_candidates) == 1
```

```
        best_sensor = sensor_candidates;
    else
        candidate_delay_sd = ...
            std_delay_ms(sensor_candidates);
        candidate_delay_sd( ...
            ~isfinite(candidate_delay_sd)) = Inf;
        [~,best_candidate_position] = ...
            min(candidate_delay_sd);
        best_sensor = ...
            sensor_candidates(best_candidate_position);
    end
end
%% 19. CREATE AND DISPLAY RESULTS TABLE
sensor_number = (1:3)';
total_fsr_peaks = total_fsr_peaks(:);
matched_beats = matched_beats(:);
matched_beat_percentage = matched_beat_percentage(:);
unmatched_fsr_peaks = unmatched_fsr_peaks(:);
unmatched_ecg_peaks = unmatched_ecg_peaks(:);
fsr_heart_rate = fsr_heart_rate(:);
heart_rate_error = heart_rate_error(:);
heart_rate_percentage_error = ...
    heart_rate_percentage_error(:);
mean_delay_ms = mean_delay_ms(:);
std_delay_ms = std_delay_ms(:);
results_table = table( ...
    sensor_number, ...
    number_of_ecg_beats*ones(3,1), ...
    total_fsr_peaks, ...
    matched_beats, ...
    matched_beat_percentage, ...
    unmatched_fsr_peaks, ...
    unmatched_ecg_peaks, ...
    ecg_heart_rate*ones(3,1), ...
    fsr_heart_rate, ...
    heart_rate_error, ...
    heart_rate_percentage_error, ...
    mean_delay_ms, ...
    std_delay_ms, ...
    'VariableNames',{ ...
    'FSR_Sensor', ...
    'ECG_Peaks', ...
    'FSR_Peaks', ...
    'Matched_Peaks', ...
    'Matched_Beat_Percent', ...
    'Unmatched_FSR_Peaks', ...
    'Unmatched_ECG_Peaks', ...
    'ECG_HR_bpm', ...
    'FSR_HR_bpm', ...
    'Absolute_HR_Error_bpm', ...
```

```
    'HR_Error_Percent', ...
    'Mean_Delay_ms', ...
    'Delay_SD_ms'});
%% Comprehensive table for the report
heart_rate_results_table = table( ...
    sensor_number, ...
    number_of_ecg_beats*ones(3,1), ...
    total_fsr_peaks, ...
    matched_beats, ...
    matched_beat_percentage, ...
    ecg_heart_rate*ones(3,1), ...
    fsr_heart_rate, ...
    heart_rate_error, ...
    heart_rate_percentage_error, ...
    mean_delay_ms, ...
    std_delay_ms, ...
    'VariableNames',{ ...
    'Sensor', ...
    'ECG_Peaks', ...
    'FSR_Peaks', ...
    'Matched_Beats', ...
    'Matched_Beat_Percent', ...
    'ECG_HR_bpm', ...
    'FSR_HR_bpm', ...
    'HR_Error_bpm', ...
    'HR_Error_Percent', ...
    'Mean_Delay_ms', ...
    'Delay_SD_ms'});
%% Display results
fprintf('\n');
fprintf('============================================================\n');
fprintf('COMPREHENSIVE FSR AND ECG RESULTS\n');
fprintf('============================================================\n');
disp(heart_rate_results_table);
fprintf('============================================================\n');
fprintf('BEST-PERFORMING FSR SENSOR\n');
fprintf('============================================================\n');
if isfinite(best_sensor)
    fprintf('Best-performing sensor: FSR Sensor %d\n', ...
        best_sensor);
    fprintf('Detected ECG R-peaks: %d\n', ...
        number_of_ecg_beats);
    fprintf('Detected FSR peaks: %d\n', ...
        total_fsr_peaks(best_sensor));
    fprintf('Matched ECG-FSR peaks: %d\n', ...
        matched_beats(best_sensor));
    fprintf('Matched-beat percentage: %.2f%%\n', ...
        matched_beat_percentage(best_sensor));
    fprintf('ECG heart rate: %.2f bpm\n', ...
        ecg_heart_rate);
```

```
    fprintf('FSR heart rate: %.2f bpm\n', ...
        fsr_heart_rate(best_sensor));
    fprintf('Absolute heart-rate error: %.2f bpm\n', ...
        heart_rate_error(best_sensor));
    fprintf('Heart-rate percentage error: %.2f%%\n', ...
        heart_rate_percentage_error(best_sensor));
    fprintf('Mean ECG-to-FSR delay: %.2f ms\n', ...
        mean_delay_ms(best_sensor));
    fprintf('Delay standard deviation: %.2f ms\n', ...
        std_delay_ms(best_sensor));
else
    fprintf('No valid FSR sensor could be selected.\n');
end
fprintf('============================================================\n');
%% 20. PLOT RAW TRIMMED SIGNALS
raw_figure = figure( ...
    'Name','Raw Trimmed Signals', ...
    'Color','w', ...
    'Position',[100 100 1400 850]);
subplot(4,1,1);
plot(time,ecg_trimmed,'k','LineWidth',1);
grid on;
ylabel('ECG');
title('Raw Signals After Synchronization and Trimming');
for sensor = 1:3
    subplot(4,1,sensor+1);
    plot( ...
        time, ...
        fsr_trimmed(:,sensor), ...
        'Color',sensorColour(sensor), ...
        'LineWidth',1);
    grid on;
    ylabel(sprintf('FSR %d',sensor));
end
xlabel('Time (s)');
%% Save raw trimmed signal figure
raw_fig_filename = fullfile( ...
    figure_output_folder, ...
    [fsr_base_name '_rawtrimmed.fig']);
raw_png_filename = fullfile( ...
    figure_output_folder, ...
    [fsr_base_name '_rawtrimmed.png']);
savefig( ...
    raw_figure, ...
    raw_fig_filename);
try
    exportgraphics( ...
        raw_figure, ...
        raw_png_filename, ...
        'Resolution',300);
```

```
catch
    print( ...
        raw_figure, ...
        raw_png_filename, ...
        '-dpng', ...
        '-r300');
end
fprintf('\nRaw trimmed figure saved as:\n');
fprintf('1. %s\n',raw_fig_filename);
fprintf('2. %s\n',raw_png_filename);
%% 21. PLOT FILTERED AND NORMALIZED SIGNALS WITH PEAKS
peak_figure = figure( ...
    'Name','Independent ECG and FSR Peak Detection', ...
    'Color','w', ...
    'Position',[100 100 1400 850]);
subplot(4,1,1);
plot(time,ecg_normalized,'k','LineWidth',1);
hold on;
plot( ...
    ecg_peak_times, ...
    ecg_peak_values, ...
    'rv', ...
    'MarkerFaceColor','r', ...
    'MarkerSize',5);
hold off;
grid on;
ylim([-0.05 1.10]);
ylabel('ECG');
title('Independent Peak Detection');
legend('ECG','R-peaks','Location','best');
for sensor = 1:3
    subplot(4,1,sensor+1);
    plot( ...
        time, ...
        fsr_normalized(:,sensor), ...
        'Color',sensorColour(sensor), ...
        'LineWidth',1);
    hold on;
    plot( ...
        fsr_peak_times{sensor}, ...
        fsr_peak_values{sensor}, ...
        'ko', ...
        'MarkerFaceColor',sensorColour(sensor), ...
        'MarkerSize',4);
    hold off;
    grid on;
    ylim([-0.05 1.10]);
    ylabel(sprintf('FSR %d',sensor));
    legend( ...
        sprintf('FSR Sensor %d',sensor), ...
```

```
        'Independent FSR peaks', ...
        'Location','best');
end
xlabel('Time (s)');
%% Save peak-detection figure
peak_fig_filename = fullfile( ...
    figure_output_folder, ...
    [fsr_base_name '_peakdetection.fig']);
peak_png_filename = fullfile( ...
    figure_output_folder, ...
    [fsr_base_name '_peakdetection.png']);
savefig( ...
    peak_figure, ...
    peak_fig_filename);
try
    exportgraphics( ...
        peak_figure, ...
        peak_png_filename, ...
        'Resolution',300);
catch
    print( ...
        peak_figure, ...
        peak_png_filename, ...
        '-dpng', ...
        '-r300');
end
fprintf('\nPeak-detection figure saved as:\n');
fprintf('1. %s\n',peak_fig_filename);
fprintf('2. %s\n',peak_png_filename);
%% 22. OVERLAY ECG WITH ALL FSR PEAKS AND MATCHED FSR PEAKS
comparison_figure = figure( ...
    'Name','ECG and FSR Peak Comparison', ...
    'Color','w', ...
    'Position',[100 100 1400 850]);
for sensor = 1:3
    subplot(3,1,sensor);
    plot( ...
        time, ...
        ecg_normalized, ...
        'k', ...
        'LineWidth',1.0, ...
        'DisplayName','Normalized ECG');
    hold on;
    plot( ...
        time, ...
        fsr_normalized(:,sensor), ...
        'Color',sensorColour(sensor), ...
        'LineWidth',1.0, ...
        'DisplayName',sprintf('Normalized FSR %d',sensor));
    plot( ...
```

```
        ecg_peak_times, ...
        ecg_normalized(ecg_peak_locations), ...
        'rv', ...
        'LineStyle','none', ...
        'MarkerFaceColor','r', ...
        'MarkerEdgeColor','r', ...
        'MarkerSize',6, ...
        'DisplayName','All ECG R-peaks');
    plot( ...
        fsr_peak_times{sensor}, ...
        fsr_peak_values{sensor}, ...
        'bo', ...
        'LineStyle','none', ...
        'MarkerFaceColor','none', ...
        'MarkerEdgeColor',[0 0.3 1], ...
        'LineWidth',1.2, ...
        'MarkerSize',6, ...
        'DisplayName','All detected FSR peaks');
    all_fsr_locations = ...
        fsr_peak_locations{sensor};
    matched_locations = ...
        matched_fsr_indices{sensor};
    is_matched_peak = ismember( ...
        all_fsr_locations, ...
        matched_locations);
    unmatched_locations = ...
        all_fsr_locations(~is_matched_peak);
    unmatched_times = ...
        time(unmatched_locations);
    unmatched_values = ...
        fsr_normalized(unmatched_locations,sensor);
    if ~isempty(unmatched_locations)
        plot( ...
            unmatched_times, ...
            unmatched_values, ...
            'x', ...
            'LineStyle','none', ...
            'Color',[1.0 0.45 0.0], ...
            'LineWidth',1.5, ...
            'MarkerSize',7, ...
            'DisplayName','Unmatched FSR peaks');
    else
        plot( ...
            NaN, ...
            NaN, ...
            'x', ...
            'LineStyle','none', ...
            'Color',[1.0 0.45 0.0], ...
            'LineWidth',1.5, ...
            'MarkerSize',7, ...
```

```
        'DisplayName','Unmatched FSR peaks');
    end
    if ~isempty(matched_fsr_indices{sensor})
        matched_fsr_values = ...
            fsr_normalized( ...
                matched_fsr_indices{sensor}, ...
                sensor);
        plot( ...
            matched_fsr_times{sensor}, ...
            matched_fsr_values, ...
            'p', ...
            'LineStyle','none', ...
            'MarkerFaceColor',[0.1 0.75 0.1], ...
            'MarkerEdgeColor',[0 0.40 0], ...
            'LineWidth',1.2, ...
            'MarkerSize',10, ...
            'DisplayName','Matched FSR peaks');
        for match_number = 1:length(matched_fsr_times{sensor})
            ecg_index = ...
                matched_ecg_indices{sensor}(match_number);
            fsr_index = ...
                matched_fsr_indices{sensor}(match_number);
            ecg_value = ...
                ecg_normalized(ecg_index);
            fsr_value = ...
                fsr_normalized(fsr_index,sensor);
            plot( ...
                [matched_ecg_times{sensor}(match_number), ...
                 matched_fsr_times{sensor}(match_number)], ...
                [ecg_value,fsr_value], ...
                ':', ...
                'Color',[0.1 0.65 0.1], ...
                'LineWidth',0.8, ...
                'HandleVisibility','off');
        end
    else
        plot( ...
            NaN, ...
            NaN, ...
            'p', ...
            'LineStyle','none', ...
            'MarkerFaceColor',[0.1 0.75 0.1], ...
            'MarkerEdgeColor',[0 0.40 0], ...
            'LineWidth',1.2, ...
            'MarkerSize',10, ...
            'DisplayName','Matched FSR peaks');
    end
    grid on;
    box on;
    ylim([-0.05 1.10]);
```

```
    xlim([time(1) time(end)]);
    ylabel('Normalized amplitude');
    title(sprintf( ...
    ['FSR Sensor %d: %d detected, %d matched ', ...
     '(%.1f%%), %d unmatched'], ...
    sensor, ...
    total_fsr_peaks(sensor), ...
    matched_beats(sensor), ...
    matched_beat_percentage(sensor), ...
    unmatched_fsr_peaks(sensor)));
    legend('Location','eastoutside');
end
xlabel('Time (s)');
sgtitle( ...
    'ECG and FSR Peak Comparison');
%% AUTOMATICALLY SAVE THE FIGURE USING THE FSR FILENAME
[fsr_folder,fsr_base_name,~] = ...
    fileparts(fsr_filename);
if isempty(fsr_folder)
    output_folder = pwd;
else
    output_folder = fsr_folder;
end
comparison_fig_filename = fullfile( ...
    output_folder, ...
    [fsr_base_name '_ecgfsr.fig']);
comparison_png_filename = fullfile( ...
    output_folder, ...
    [fsr_base_name '_ecgfsr.png']);
%% Save editable MATLAB figure
savefig( ...
    comparison_figure, ...
    comparison_fig_filename);
%% Save high-resolution PNG image
try
    exportgraphics( ...
        comparison_figure, ...
        comparison_png_filename, ...
        'Resolution',300);
catch
    print( ...
        comparison_figure, ...
        comparison_png_filename, ...
        '-dpng', ...
        '-r300');
end
%% Display saved filenames
fprintf('\n============================================================\n');
fprintf('ECG-FSR COMPARISON FIGURE SAVED\n');
fprintf('============================================================\n');
```

```
fprintf('MATLAB figure: %s\n',comparison_fig_filename);
fprintf('PNG image: %s\n',comparison_png_filename);
fprintf('============================================================\n');
%% 25. SAVE ANALYSIS RESULTS AUTOMATICALLY
analysis_mat_filename = fullfile( ...
    figure_output_folder, ...
    [fsr_base_name '_analysis.mat']);
analysis_csv_filename = fullfile( ...
    figure_output_folder, ...
    [fsr_base_name '_analysis.csv']);
%% Save all MATLAB analysis variables
save( ...
    analysis_mat_filename, ...
    'time', ...
    'fsr_raw', ...
    'fsr_trimmed', ...
    'ecg_trimmed', ...
    'fsr_filtered', ...
    'ecg_filtered', ...
    'fsr_normalized', ...
    'ecg_normalized', ...
    'fsr_peak_locations', ...
    'fsr_peak_times', ...
    'ecg_peak_locations', ...
    'ecg_peak_times', ...
    'matched_ecg_times', ...
    'matched_fsr_times', ...
    'matched_ecg_indices', ...
    'matched_fsr_indices', ...
    'delay_seconds', ...
    'number_of_ecg_beats', ...
    'total_fsr_peaks', ...
    'matched_beats', ...
    'matched_beat_percentage', ...
    'unmatched_fsr_peaks', ...
    'unmatched_ecg_peaks', ...
    'fsr_heart_rate', ...
    'ecg_heart_rate', ...
    'heart_rate_error', ...
    'heart_rate_percentage_error', ...
    'mean_delay_ms', ...
    'std_delay_ms', ...
    'results_table', ...
    'heart_rate_results_table', ...
    'best_sensor', ...
    'analysis_duration');
%% Save the results table as CSV
writetable( ...
    results_table, ...
    analysis_csv_filename);
```

```
%% Display saved filenames
fprintf('\n============================================================\n');
fprintf('ANALYSIS FILES SAVED SUCCESSFULLY\n');
fprintf('============================================================\n');
fprintf('MATLAB result: %s\n',analysis_mat_filename);
fprintf('CSV result: %s\n',analysis_csv_filename);
%% LOCAL FUNCTION: MIN-MAX NORMALIZATION
function normalized_signal = normalizeMinMax(input_signal)
   minimum_value = min(input_signal);
   maximum_value = max(input_signal);
   signal_range = ...
      maximum_value-minimum_value;
   if signal_range <= eps
      normalized_signal = ...
         zeros(size(input_signal));
   else
      normalized_signal = ...
         (input_signal-minimum_value)/signal_range;
   end
end
%% LOCAL FUNCTION: SENSOR COLOURS
function colour = sensorColour(sensor_number)
   colours = [
      0.0000 0.4470 0.7410
      0.8500 0.3250 0.0980
      0.4660 0.6740 0.1880
   ];
   colour = colours(sensor_number,:);
end
```

APPENDIX C

Listing C.1 Arduino Uno Code for Three-Channel FSR Data Acquisition

```
// Three-channel FSR data acquisition
const int fsrPins[] = {A0, A1, A2};
const int numSensors = sizeof(fsrPins) / sizeof(fsrPins[0]);

const unsigned long interval = 10;
unsigned long previousMillis = 0;

void setup() {
  Serial.begin(115200);
}

void loop() {
  unsigned long currentMillis = millis();

  if (currentMillis - previousMillis >= interval) {
    previousMillis = currentMillis;

    for (int i = 0; i < numSensors; i++) {
      int fsrReading = analogRead(fsrPins[i]);
      Serial.print(fsrReading);

      if (i < numSensors - 1) {
        Serial.print("\t");
      }
    }

    Serial.println();
  }
}
```